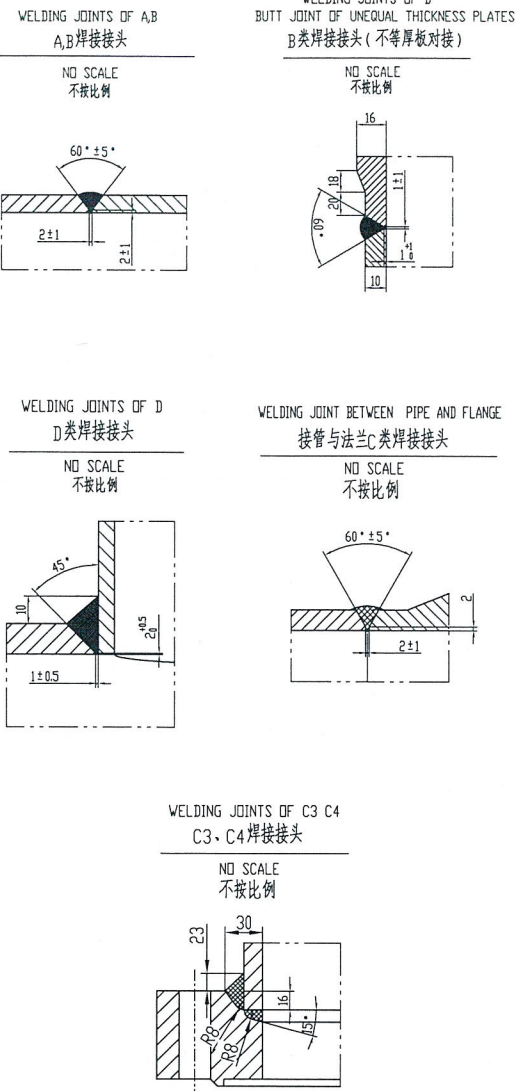
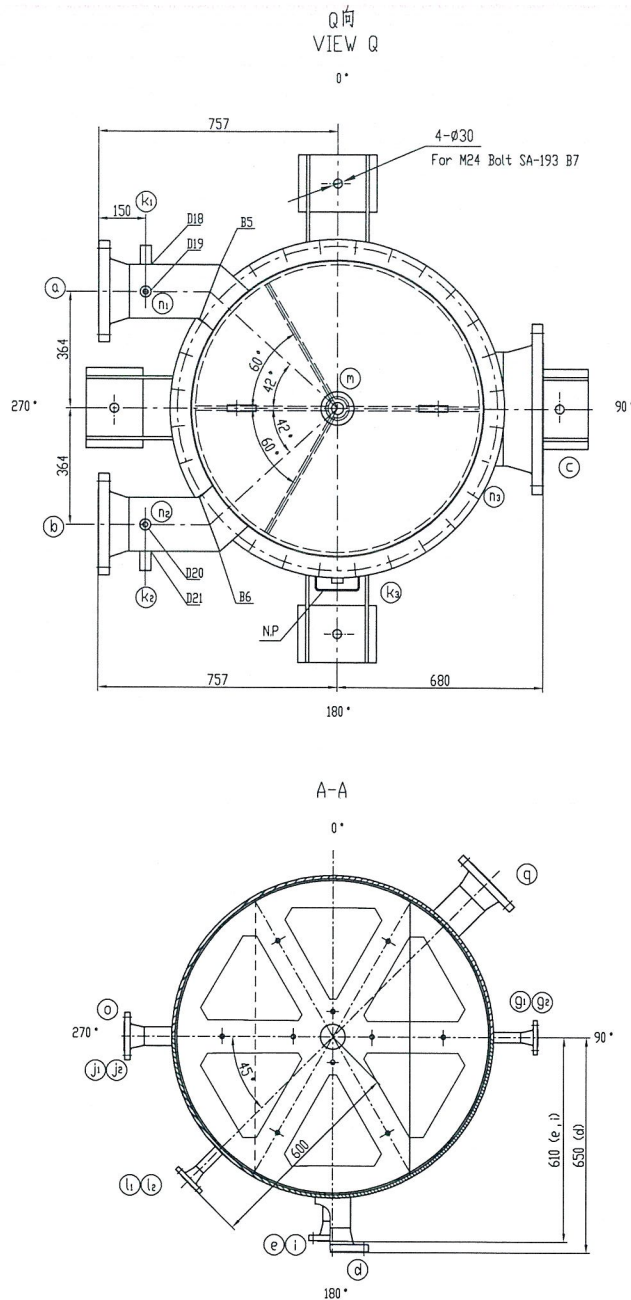
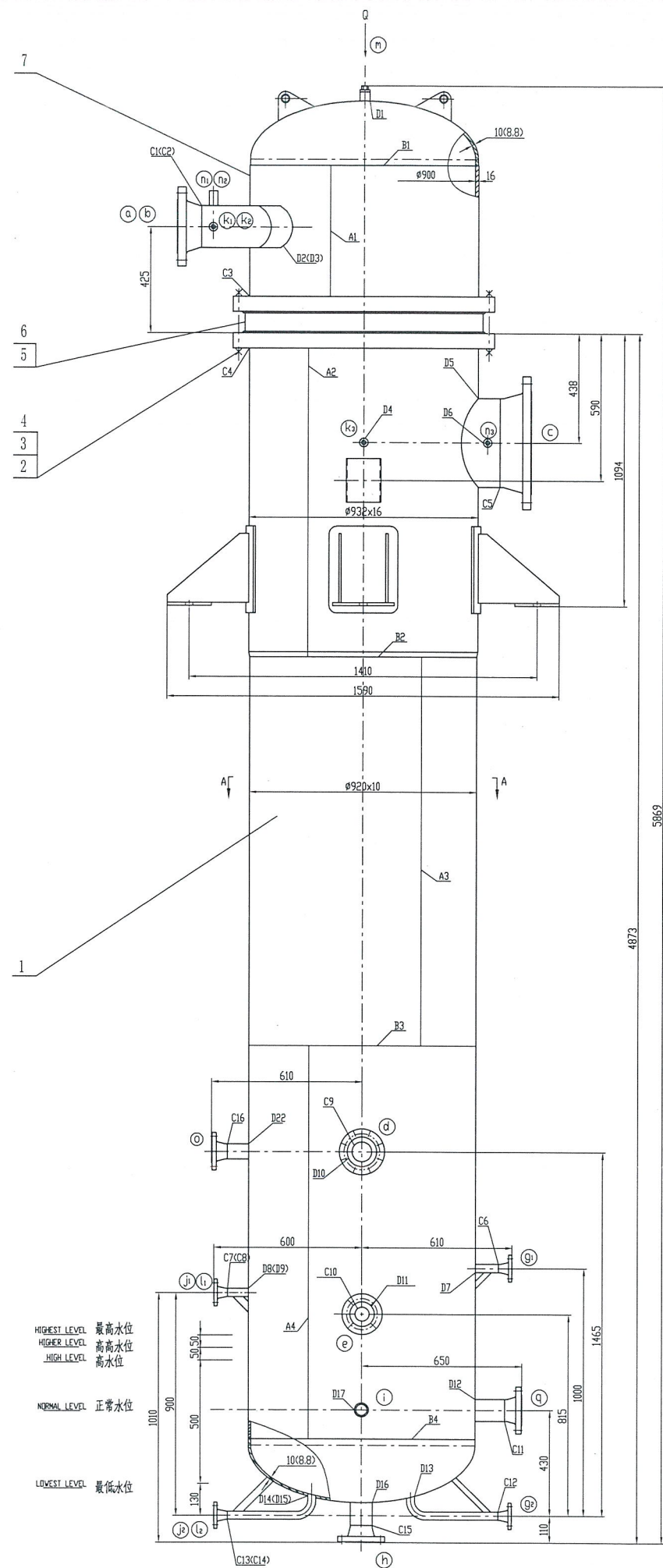




- TECHNICAL REQUIREMENTS
1. THE PERPENDICULARITY TOLERANCE OF TUBESHEET SEAL SURFACE AND SHELL AXIS IS 2mm.
 2. TUBE-TO-TUBESHEET JOINT IS STRENGTH WELDING.
 3. LIFTING LUG IN WATER CHAMBER HEAD IS ONLY USED TO LIFT THE WATER CHAMBER, NOT THE WHOLE EQUIPMENT.
 4. AFTER ASSEMBLY OF SHELL AND TUBE BUNDLES, PERFORM HYDROTEST AT SHELL SIDE. THE TUBE END SHOULD BE FREE OF LEAKAGE. AFTER THE TEST IS ACCEPTED, PERFORM HYDROTEST AT TUBE SIDE.
 5. PERFORM MT TO WELDING OF THE BACKING PLATE OF SADDLE, IN ACCORDANCE WITH ASME VIII DIV.1 APPENDIX 6.
 6. BEFORE DELIVERY, CAP EACH NOZZLE. FILL IN NITROGEN OF 0.15MPa. THE NITROGEN FILLING PRESSURE SHALL NOT BE MORE THAN 0.1MPa BEFORE DELIVERY.
 7. SAFETY ACCESSORIES ARE INSTALLED ON THE PIPELINE.
- 技术要求
1. 管板密封面与壳体轴线垂直度公差为2mm。
 2. 管子与管板的连接采用强度焊。
 3. 水箱封头上的吊耳只用于起吊水箱，不能用于起吊整个设备。
 4. 管系与壳体组装完成后进行水压试验，并检查管头不得渗漏，最后进行管程水压试验。
 5. 耳式支座垫板与筒体的焊缝进行100%磁粉检测，符合ASME第V卷第1册附录6要求。
 6. 设备出厂前各管口须封盖，并充氮保护，充氮压力为0.15MPa，出厂前不得低于0.1MPa。
 7. 安全附件设置在管路上。

TÜV Rheinland
inspected reviewed witnessed
2024.6.18

DESIGN SPECIFICATION 设计数据表									
DESIGN AND MANUFACTURE CODE 设计和制造规范	ASME CODE SECTION VIII, DIV.1, 2023.0 工厂或工厂制造规范	ASME CODE SECTION VIII, DIV.1, 2023.0 工厂或工厂制造规范	ASME CODE SECTION VIII, DIV.1, 2023.0 工厂或工厂制造规范	ASME CODE SECTION VIII, DIV.1, 2023.0 工厂或工厂制造规范	ASME CODE SECTION VIII, DIV.1, 2023.0 工厂或工厂制造规范	ASME CODE SECTION VIII, DIV.1, 2023.0 工厂或工厂制造规范	ASME CODE SECTION VIII, DIV.1, 2023.0 工厂或工厂制造规范	ASME CODE SECTION VIII, DIV.1, 2023.0 工厂或工厂制造规范	ASME CODE SECTION VIII, DIV.1, 2023.0 工厂或工厂制造规范
OPERATING PRESSURE 工作压力 MPa.G	0.0718 (G)	0.0718 (G)	0.0718 (G)	0.0718 (G)	0.0718 (G)	0.0718 (G)	0.0718 (G)	0.0718 (G)	0.0718 (G)
DESIGN PRESSURE 设计压力 MPa.G	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
MAWP 最大允许工作压力 MPa.G	0.1899	0.1899	0.1899	0.1899	0.1899	0.1899	0.1899	0.1899	0.1899
OPERATING TEMPERATURE 工作温度 °C	89.97	89.97	89.97	89.97	89.97	89.97	89.97	89.97	89.97
DESIGN TEMPERATURE 设计温度 °C	99	99	99	99	99	99	99	99	99
最低设计金属温度 °C	0°C AT 0.1 MPa	0°C AT 0.1 MPa	0°C AT 0.1 MPa	0°C AT 0.1 MPa	0°C AT 0.1 MPa	0°C AT 0.1 MPa	0°C AT 0.1 MPa	0°C AT 0.1 MPa	0°C AT 0.1 MPa
MEDIUM (IN/OUT) 介质	STEAM/WATER 蒸汽/水	STEAM/WATER 蒸汽/水	STEAM/WATER 蒸汽/水	STEAM/WATER 蒸汽/水	STEAM/WATER 蒸汽/水	STEAM/WATER 蒸汽/水	STEAM/WATER 蒸汽/水	STEAM/WATER 蒸汽/水	STEAM/WATER 蒸汽/水
MEDIUM CHARACTERISTIC 介质特性	NONE-LETHAL 非致命	NONE-LETHAL 非致命	NONE-LETHAL 非致命	NONE-LETHAL 非致命	NONE-LETHAL 非致命	NONE-LETHAL 非致命	NONE-LETHAL 非致命	NONE-LETHAL 非致命	NONE-LETHAL 非致命
CORROSION ALLOWANCE 腐蚀裕量	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
JOINT COEFFICIENT 焊接接头系数	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
TOTAL VOLUME 总体积 m³	2.34	2.34	2.34	2.34	2.34	2.34	2.34	2.34	2.34
HEAT TRANSFER AREA 传热面积 m²	140	140	140	140	140	140	140	140	140
TUBE DIMENSION 管子规格 mm	ø16x0.9/1.2	ø16x0.9/1.2	ø16x0.9/1.2	ø16x0.9/1.2	ø16x0.9/1.2	ø16x0.9/1.2	ø16x0.9/1.2	ø16x0.9/1.2	ø16x0.9/1.2
NO. OF TUBE PASSES 管子程数	1	1	1	1	1	1	1	1	1
IMPACT TEST 冲击试验	NOT REQUIRED 无需冲击试验	NOT REQUIRED 无需冲击试验	NOT REQUIRED 无需冲击试验	NOT REQUIRED 无需冲击试验	NOT REQUIRED 无需冲击试验	NOT REQUIRED 无需冲击试验	NOT REQUIRED 无需冲击试验	NOT REQUIRED 无需冲击试验	NOT REQUIRED 无需冲击试验
SAFETY VALVE ARRANGEMENT 安全阀布置	-	-	-	-	-	-	-	-	-
SET PRESSURE OF SAFETY VALVE 安全阀设定压力 MPa	-	-	-	-	-	-	-	-	-
HYDRAULIC TEST PRESSURE 水压试验压力 MPa.G	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
HYDRAULIC TEST TEMPERATURE 水压试验温度 °C	10	10	10	10	10	10	10	10	10
GAS LEAKAGE TEST PRESSURE 气密性试验压力 bar.G	-	-	-	-	-	-	-	-	-
NOZZLE TABLE 管口表									
NOZZLE 管口	CLASS 压力等级	NPS 公称管径	FLANGE CODE 法兰标准	CONT. TYPE 连接形式	USE 用途	REMARKS 备注			
a	300	DN150 6"	ASME B16.5-2020	WN RF	FEEDWATER INLET 给水进口				
b	300	DN150 6"	ASME B16.5-2020	WN RF	FEEDWATER OUTLET 给水出口				
c	150	DN350 14"	ASME B16.5-2020	WN RF	STEAM INLET 蒸汽进口				
d	150	DN80 3"	ASME B16.5-2020	WN RF	DRAIN INLET NOZZLE 旁路排水入口				
e	150	DN50 2"	ASME B16.5-2020	WN RF	VENT NOZZLE AT SHELL SIDE 壳体排气口				
g1-2	150	DN25 1"	ASME B16.5-2020	WN RF	LEVEL GAGE NOZZLE 液位计接口				
h	150	DN80 3"	ASME B16.5-2020	WN RF	DRAIN NOZZLE 排水口				
i	150	DN50 2"	ASME B16.5-2020	WN RF	STEAM NOZZLE 蒸汽接口				
j1-2	150	DN25 1"	ASME B16.5-2020	WN RF	MAGNETIC LEVEL GAGE NOZZLE 磁翻板液位计接口				
k1-2	150	M20x1.5	ASME B16.5-2020	WN RF	PRESSURE GAGE NOZZLE 压力表接口				
l1-3	150	DN25 1"	ASME B16.5-2020	WN RF	REMOTE TRANSDUCER MAGNETIC FLIP LEVEL GAGE 远传磁翻板液位计接口				
m	150	M24x1.5	ASME B16.5-2020	WN RF	VENT NOZZLE OF WATER CHAMBER 水箱排气口				
n1-3	150	M27x2	ASME B16.5-2020	WN RF	TEMPERATURE NOZZLE 温度计接口				
o	150	DN50 2"	ASME B16.5-2020	WN RF	RC NOZZLE 充氮保护接口				
q	150	DN80 3"	ASME B16.5-2020	WN RF	EMERGENCY DRAIN NOZZLE 紧急排水口				
THE MATERIAL SPECIFICATION INCLUDING THE GRADE, TYPE, AND CLASS IF APPLICABLE OF MAIN PRESSURE PARTS									
主要承压元件材料标准，包括等级、类型、级别（如果有的话）									
THE ELEMENT 元件名称	THE SUBASSEMBLY 元件所属部件	THE MATERIAL SPECIFICATION 材料标准	THE ELEMENT 元件名称	THE SUBASSEMBLY 元件所属部件	THE MATERIAL SPECIFICATION 材料标准				
ELLIPTICAL HEAD 椭圆形封头	WATER CHAMBER 水箱	SA-516M Gr.485	CYLINDER 筒体	WATER CHAMBER 水箱	SA-516M Gr.485				
SHELL OF WATER CHAMBER 水箱筒体	WATER CHAMBER 水箱	SA-516M Gr.485	ELLIPTICAL HEAD 椭圆形封头	WATER CHAMBER 水箱	SA-516M Gr.485				
NOZZLE 接管	WATER CHAMBER 水箱	SA-106M Gr.B	NOZZLE 接管	WATER CHAMBER 水箱	SA-106M Gr.B				
TUBESHEET 管板	TUBE BUNDLE 管束	SA-266M Gr.2	NOZZLE FLANGE 接管法兰	WATER CHAMBER 水箱	SA-105M				
U-TUBE U型管	TUBE BUNDLE 管束	SA-688M TP304							
件号 图号或标准									
ITEM	DRAW NO.	DESCRIPTION	QTY.	MATERIAL	UNIT	WEIGHT (kg)	NOTE		
7	TXNX23-021.02	SHELL 筒体	1	COMPONENT PARTS 组合件	1540				
6	TXNX23-020.04	GASKET OF SHELL SIDE 壳体垫片	1	PTFE	0.1		REFER TO 参照		
5	TXNX23-021.01	TUBE BUNDLE 管束	1	COMPONENT PARTS 组合件	1705				
4	TXNX23-020.02	GASKET OF WATER SIDE 管束垫片	1	PTFE	0.2		REFER TO 参照		
3	ASME B18.2.2	NUTS 螺母 M24	56	SA-194M 2H	0.05	2.4			
2	ASME B18.2.1	FULL SCREW BOLTS 全螺纹螺栓 M24x270	28	SA-193M B7	0.73	17.52			
1	TXNX23-020.01	WATER CHAMBER 水箱	1	COMPONENT PARTS 组合件	573		REFER TO 参照		
LOW PRESSURE HEATER JD-140				TXNX23-021.00 A23-0021 Z241.82.62-1					
低压机热器				RENEWABLE ENERGY POWER PLANT PHASE 1 PROJECT IN SELANGOR STATE, MALAYSIA 马来西亚雪兰莪州再生能源电站一期项目					
JD-140				南京汽轮电机集团泰兴宁兴机械有限公司 NANJING TURBOELECTRIC MACHINERY GROUP TAIXING NINGXING MACHINERY CO., LTD.					

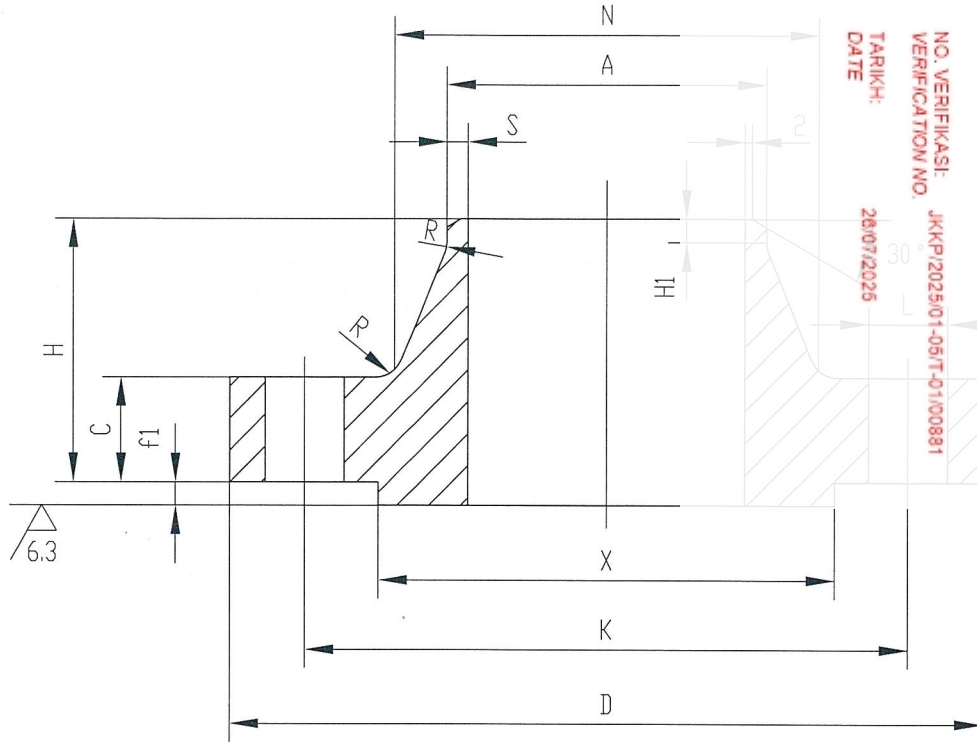


JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



Verifikasi Dokumen
Document Verification

NO. VERIFIKASI: JKKP/2025/01-06/T-01/00881
VERIFICATION NO.
TARIKH: 28/07/2025
DATE



TÜVRheinland®

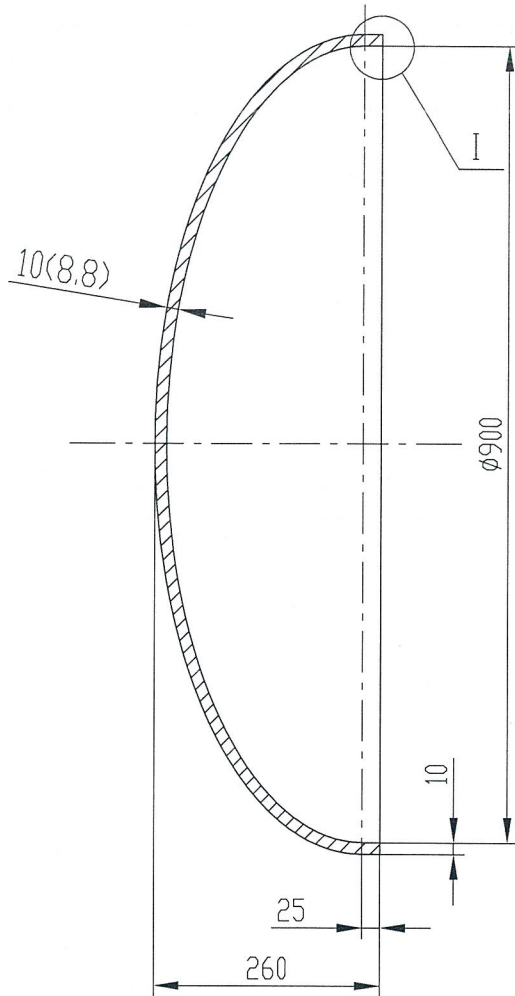
☐ Inspected ☒ Reviewed ☐ Witnessed

2024.6.18

ASME B16.5 WN RF

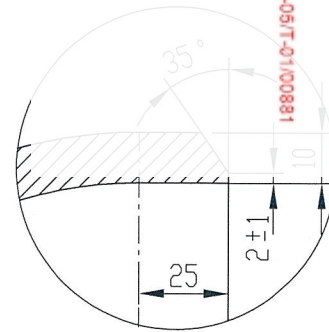
MODEL 型号	A	D	K	L	NL OF SCREW HOLES 螺孔数	X	f1	C	H	S	N	R	NL OF FLANGE 法兰数	H1	UNIT WEIGHT 单件重	MATERIAL 材料
class300 6"	168.3	318	269.7	7/8"	12	215.9	2	35	97	10.97	206	13	2	16	12	SA-105M

						FLANGE 法兰				TXNX23-020.01.01	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
DESIGNED BY 设计	2020.6.14										
CHECKED BY 校对	2024.6.4										
REVIEWED BY 审核											
APPROVED BY 批准						PAGE OF 共 张 第 张					



DETAIL DRAWING OF SECTION

I 部详图



NOTES

技术要求

- 1.THE PLATE USED FOR HEAD SHALL BE SUPPLIED IN A ROLLED STATE AND SHALL COMPLY WITH SA-516 SPECIFICATION IN PART A OF ASME SECTION II; 封头用板材以轧制状态供货, 且应符合ASME第II卷A篇SA-516规范的所有相关要求;
- 2.THIS HEAD IS A 2:1 STANDARD ELLIPTICAL HEAD AND SHALL BE MANUFACTURED IN ACCORDANCE WITH ASME SECTION VIII; 此封头为2:1的标准椭圆封头, 应按照ASME第VIII卷有关各篇制造方面的要求制造;
- 3.THE MINIMUM THICKNESS AFTER HEAD FORMING SHALL NOT BE LESS THAN 8.8MM. 封头成形后最小厚度不得小于8.8MM.

TÜVRheinland
☐ inspected ☒ reviewed ☐ witnessed
 2024.6.14

						HEAD 封头 900x10(8.8)				TXNX23-020.01.05	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日						
DESIGNED BY 设计	陈伟		2024.6.14			MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
CHECKED BY 校对	陈伟		2024.6.14			SA-516M.Gr.485	1	74.1			
REVIEWED BY 审核											
APPROVED BY 批准											
						PAGE OF 共 张 第 张					

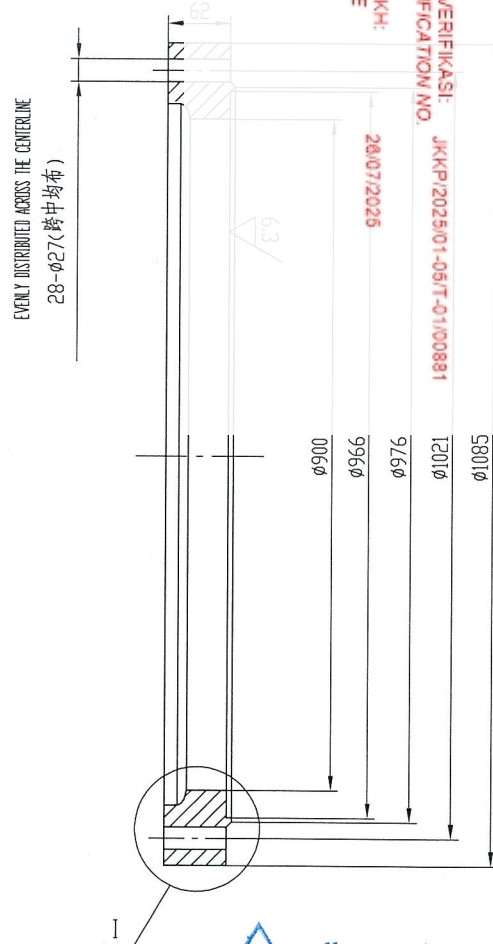


JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



Verifikasi Dokumen
Document Verification

NO. VERIFIKASI: JKKP/2025/01-06/IT-01/00881
VERIFICATION NO.
TARIKH: 26/07/2025
DATE



TÜVRheinland®
☐ inspected ☒ reviewed ☐ witnessed
2024.6.18

NOTES
技术要求

1.THE FLANGE MATERIAL SHALL BE SUPPLIED IN A NORMALIZED STATE
AND SHALL COMPLY WITH ASME SECTION II, PART A.
法兰材料以正火状态供货，且应符合ASME第II卷A篇SA-105M规范的所有相关要求。

						GIRTH FLANGE 设备法兰				TXNX23-020.01.07					
														TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日										
DESIGNED BY 设计		2024.6.14				MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例						
CHECKED BY 校对		2024.6.14				SA-105M	1	110	1:2						
REVIEWED BY 审核						PAGE OF 共 张 第 张									
APPROVED BY 批准															

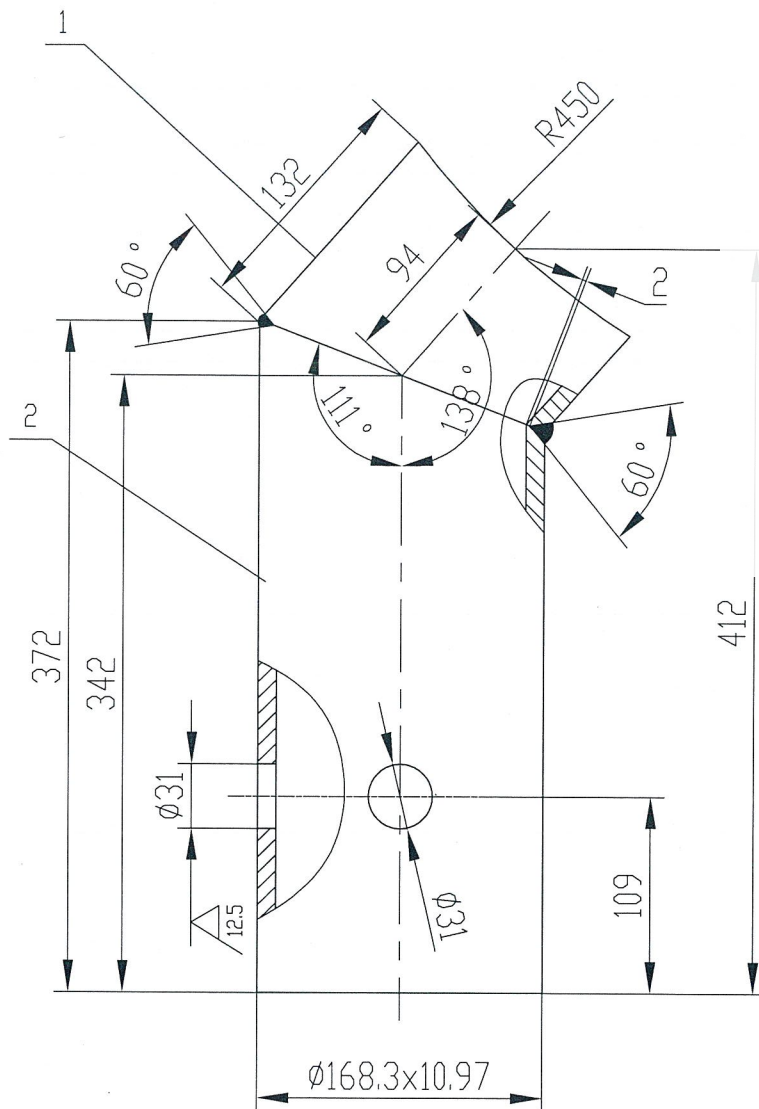
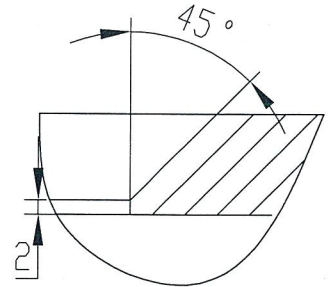


Verifikasi Dokumen
Document Verification

NO. VERIFIKASI: JK/P/2025/01-05/T-01/00881
TARIKH: 26/07/2025
DATE

DRAWING OF OPENING GROOVE
开孔坡口图

1:1



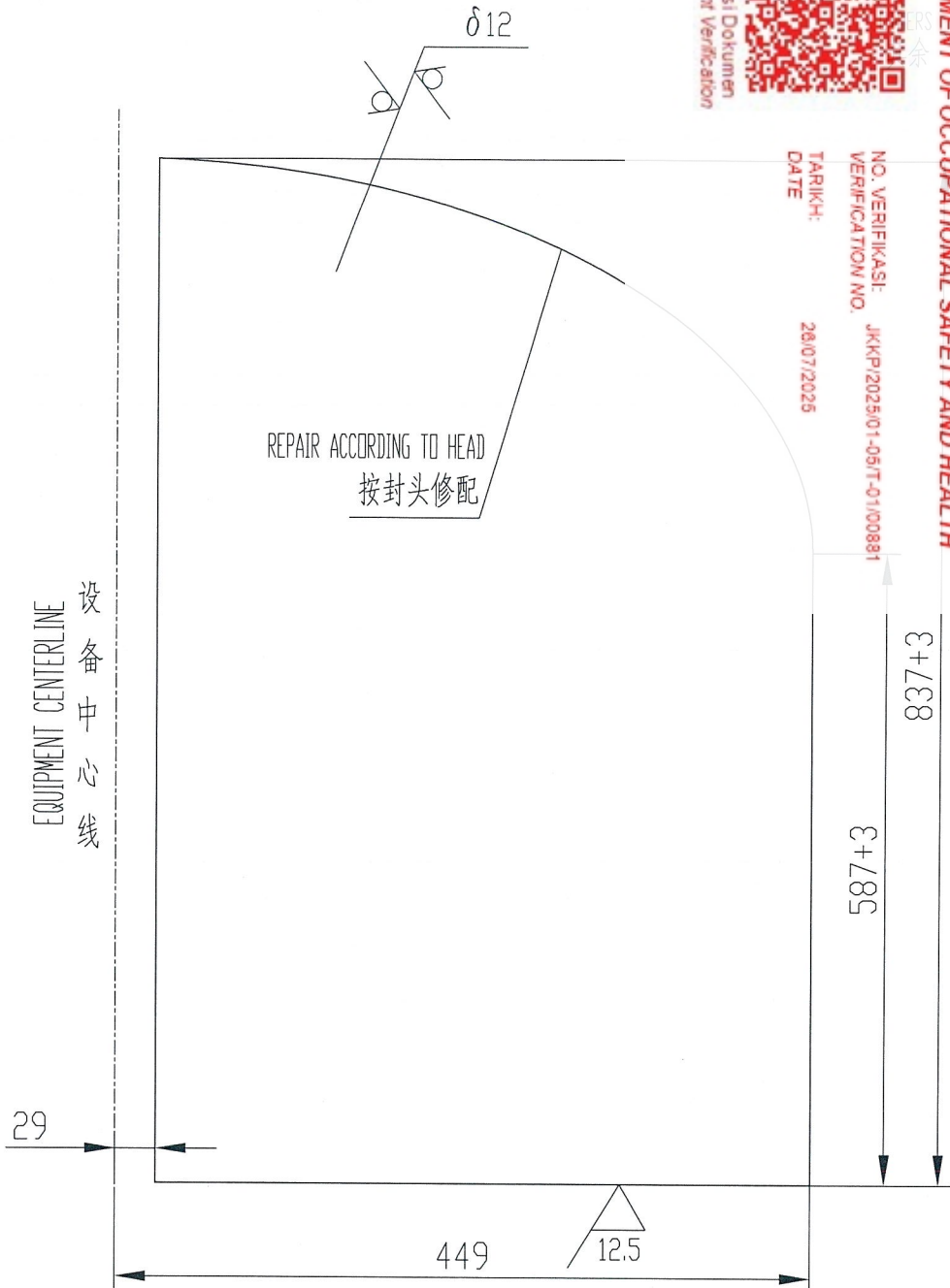
2		STEEL PIPE 钢管Ø168.3X10.97	1	SA-106M Gr.B		12.5	
1		STEEL PIPE 钢管Ø168.3X10.97	1	SA-106M Gr.B		3.2	
NO. 序号	DRAWING NO. 代号	DESCRIPTION 名称	QTY 数量	MATERIAL 材料	UNIT 单重	TOTAL 总重	NOTE 备注
					WEIGHT 重量		
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日		
DESIGNED BY 设计	陈伟		2024.6.14		MATERIAL 材料	QTY 数量	WEIGHT 重量
CHECKED BY 校对	侯好		2024.6.14			1	15.7
REVIEWED BY 审核							1:4
APPROVED BY 批准					TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司		
INLE PIPE 进水管							
TXNX23-020.01.08							
PAGE OF 共 张 第 张							



JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



NO. VERIFIKASI: JKKP/2025/01-06/T-01/00881
VERIFICATION NO.
TARIKH: 26/07/2025
DATE



TÜVRheinland®

☐ inspected ☒ reviewed ☐ witnessed

2024.6.18

						PASS PARTITION 分程隔板				TXNX23-020.01.09			
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日								
DESIGNED BY 设计	陈伟		2024.6.14			MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司			
CHECKED BY 校对	徐子		2024.6.14			SA-516M Gr.485	4	32	1:5				
REVIEWED BY 审核						PAGE OF 共 张 第 张							
APPROVED BY 批准													



NO. VERIFIKASI: JKPP/2025/01-05/IT-01/00881
VERIFICATION NO.
TARİKHİ: 28/07/2025
DATE



45.

☐ inspected ☒ reviewed ☐ witnessed

2024. 6-18

2		STEEL PIPE 钢管φ168.3X10.97	1	SA-106M Gr.B		12.5	
1		STEEL PIPE 钢管φ168.3X10.97	1	SA-106M Gr.B		3.2	
NO. 序号	DRAWING NO. 代号	DESCRIPTION 名 称	QTY 数量	MATERIAL 材料	UNIT 单重	TOTAL 总重	NOTE 备注
					WEIGHT 重 量		
					TXNX23-020.01.10 		
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料 QTY 数量 WEIGHT 重量 PROPORTION 比例 	
DESIGNED BY 设计	陈伟		2024.6.14				
CHECKED BY 校对	陈伟		2024.6.14				
REVIEWED BY 审核							
APPROVED BY 批准							
PAGE OF 共 张 第 张						TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	

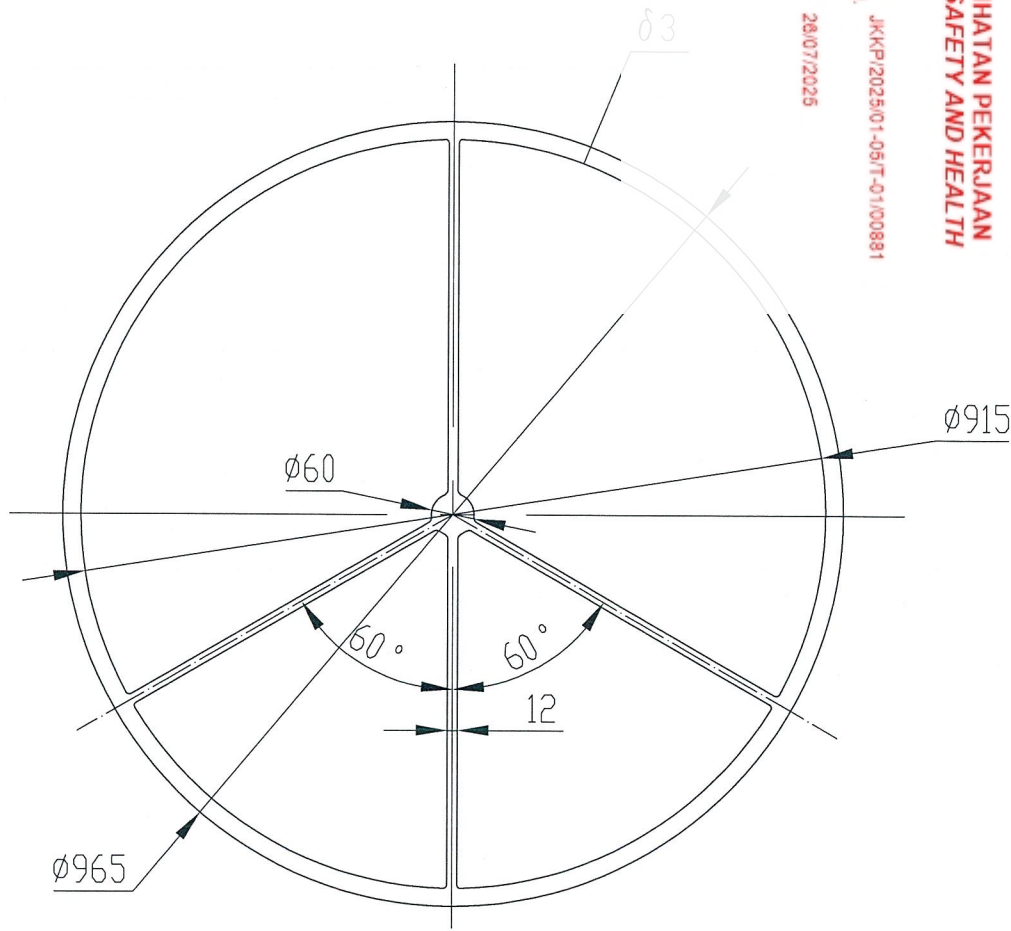


Verifikasi Dokumen
Document Verification

JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



NO. VERIFIKASI: JKKP/2025/01-06/IT-01/00881
VERIFICATION NO.
TARIKH: 26/07/2025
DATE

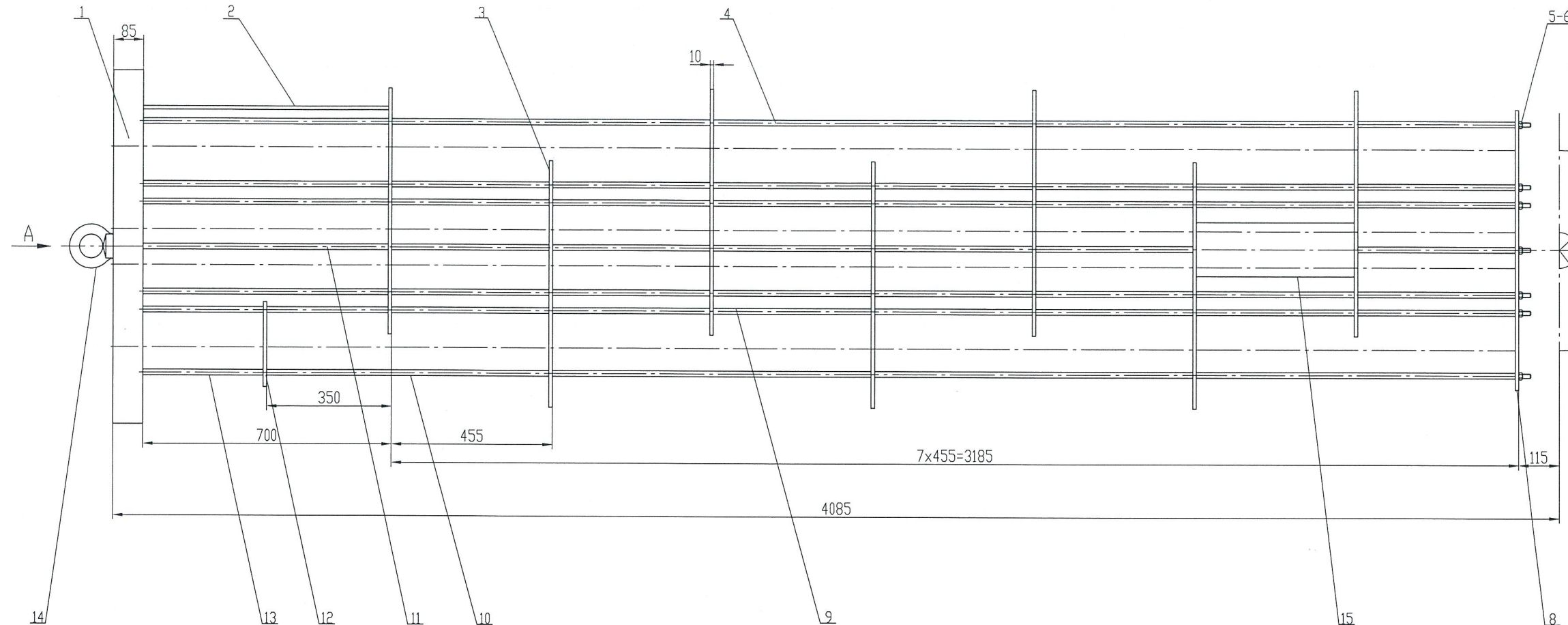


TÜVRheinland®
☐ inspected ☒ reviewed ☐ witnessed
Li Jia
2024.6.18

						WATER SIDE GASKET 水侧垫片				TXNX23-020.02	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	 南京汽轮机集团 泰兴宁兴机械有限公司 NANJING TURBINE&ELECTRIC MACHINERY GROUP TAIXING NINGXING MACHINERY CO., LTD.	
DESIGNED BY 设计	陈伟		2024.6.14		PTFE	1	0.2	1:10			
CHECKED BY 校对	陈伟		2024.6.14								
REVIEWED BY 审核											
APPROVED BY 批准					PAGE OF 共 张 第 张						



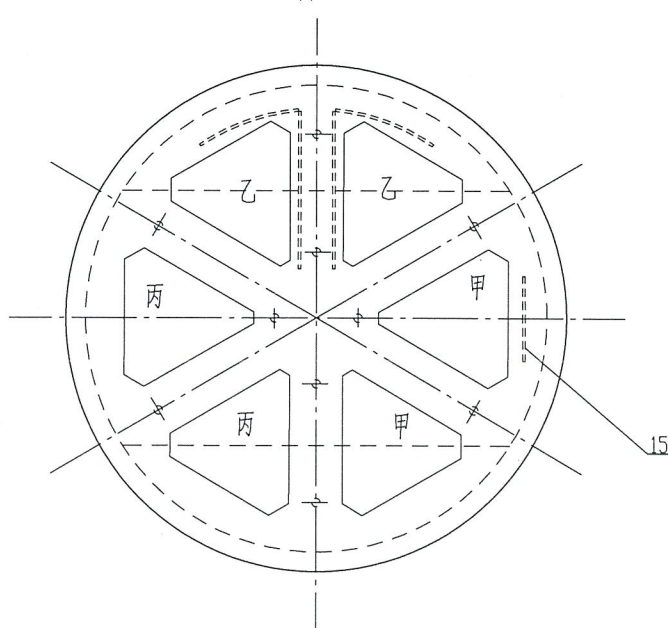
NO. VERIFIKASI: JKKP/2025/01-05/TT-01/00881
VERIFICATION NO.
TARIKH: 25/07/2025
DATE



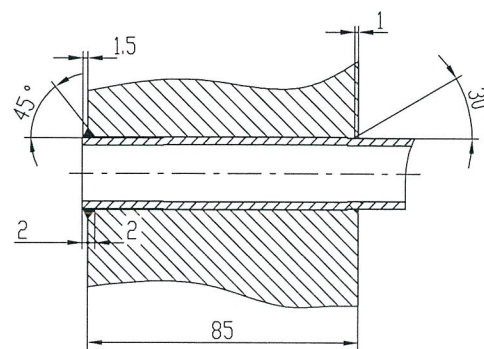
NOTES
技术要求

- 1.TUBE BUNDLES SHALL BE MANUFACTURED IN ACCORDANCE WITH ASME SECTION VIII;
管束应按照ASME 第Ⅷ卷有关篇制造方面的要求制造;
- 2.TUBE AND TUBE PLATE ARE CONNECTED BY WELDING;
换热管与管板采用焊接的方式连接;
- 3.TIGHTEN THE TIE ROD NUTS AND SPOT WELD THEM FIRMLY;
拉杆螺母旋紧后点焊牢固;
- 4.AFTER COMPLETION OF TUBE BUNDLES, IT IS NECESSARY TO REMOVE IRON FILINGS AND DEBRIS INSIDE AND OUTSIDE TUBES, AS WELL AS OTHER PARTS OF TUBE BUNDLES;
管系完工后需清除管内外及管系其他部分铁屑杂物;
- 5.FOR OTHER TECHNICAL REQUIREMENTS, REFER TO GA DRAWING.
其他技术要求见总图。

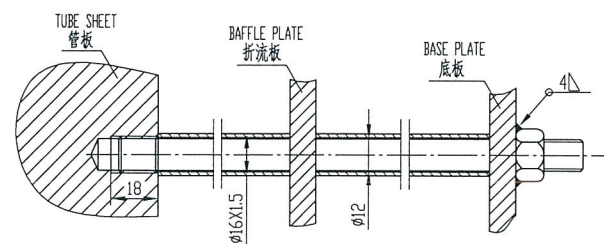
VIEW A
A向



DETAIL DRAWING OF TUBE WELDING
换热管焊接详图



TIE ROD DRAWING
拉杆结构图



TÜVRheinland®

inspected reviewed witnessed

2024.6.18

15		DRAIN INLET BAFFLE 疏水入口折板 445x150x6	1	S30408		3.2	
14	GB/T 825-1988	SCREW 螺栓M36	1	35		2	
13		SPACER 定距管 $\phi 16 \times 1.5$ l=340	5	20	0.18	0.9	
12	TXNX23-021.01.04	PASS PARTITION 隔板	1	Q235B		12	
11		SPACER 定距管 $\phi 16 \times 1.5$ l=690	7	20	0.37	2.59	
10		SPACER 定距管 $\phi 16 \times 1.5$ l=795	1	20		0.43	
9		SPACER 定距管 $\phi 16 \times 1.5$ l=445	57	20	0.23	13.11	
8	TXNX23-020.03.05	BASE PLATE 底板	1	Q235B		30	BORROW 借用
7	TXNX23-021.01.03	U-TUBE U形管	339	SA-688M TP304		985	
6	GB/T 6170	NUT 螺母M12	10	8	0.01	0.1	
5	TXNX23-021.01.02	TIE ROD 拉杆	10	Q235B	3.48	34.8	
4		SPACER 定距管 $\phi 16 \times 1.5$ l=900	6	20	0.48	2.88	
3	TXNX23-020.03.02	BAFFLE PLATE 折流板	7	Q235B	37	259	BORROW 借用
2	TXNX23-021.01.01	INLET BAFFLE 进口折板	1	SA-240M 304		4.84	
1	TXNX23-020.03.01	TUBE SHEET 管板	1	SA-266M Gr.2		357	BORROW 借用
NOL 序号	DRAWING NO. OR STANDARD 图号或标准	DESCRIPTION 名称	QTY 数量	MATERIAL 材料	UNIT 单位	TOTAL 重量	NOTE 备注

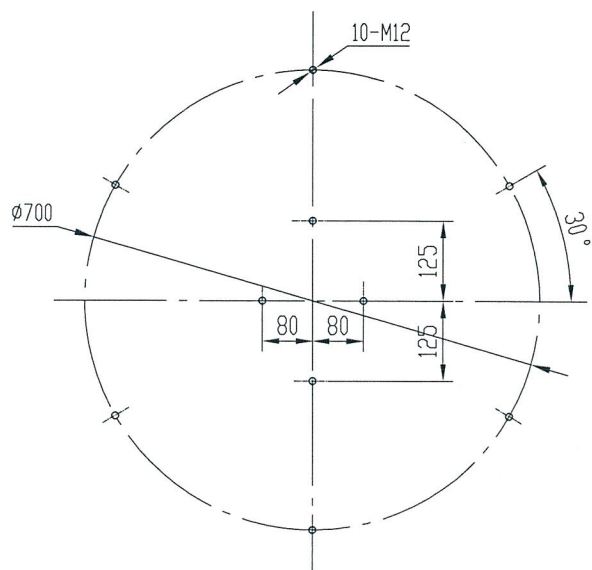
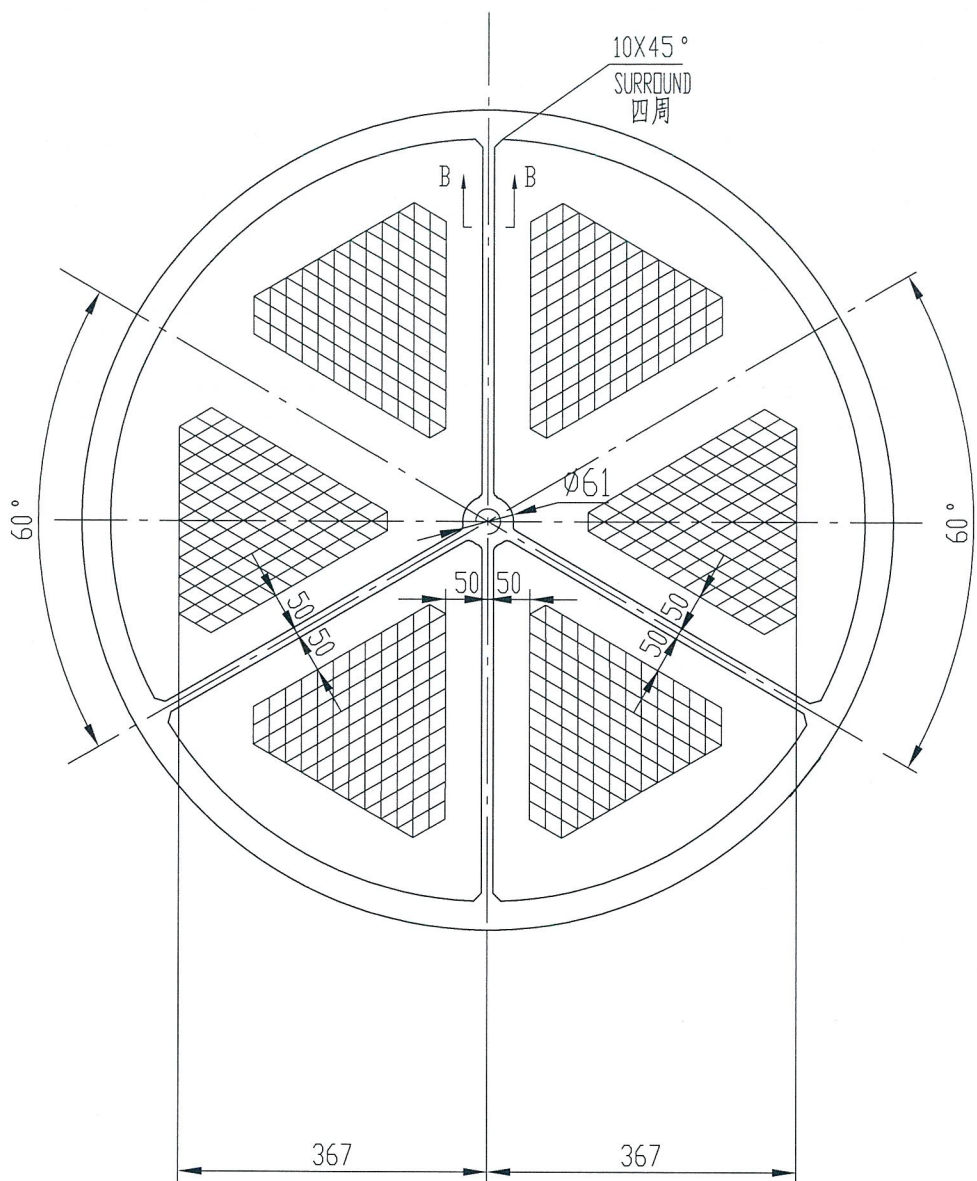
TUBE BUNDLES 管束				TXNX23-021.01			
NO. 序号	QTY 数量	REVISION 变更	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例
1	1	2024.6.14	2024.6.14	SA-266M Gr.2	1	1705	1:7
PAGE OF 共 1 页				南京汽轮机集团有限公司 NANJING TURBOELECTRIC MACHINERY GROUP TAIXING NINGXING MACHINERY CO., LTD.			

DATE

28/07/2025

管板完工后须清除所有表面的毛刺、铁屑等杂物。

TIE ROD POSITION AT VIEW A
A向拉杆孔位置



Lu Jian
witnessed
20 Feb. 18

						TUBE SHEET 管板				TXNX23-020.03.01			
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日								
DESIGNED BY 设计	陈伟		2024.6.14			MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司			
CHECKED BY 校对	段明		2024.6.14			SA-266M Gr.2	1	357	1:5				
REVIEWED BY 审核													
APPROVED BY 批准						PAGE OF 共 张 第 张							

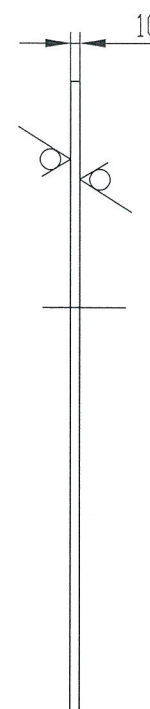
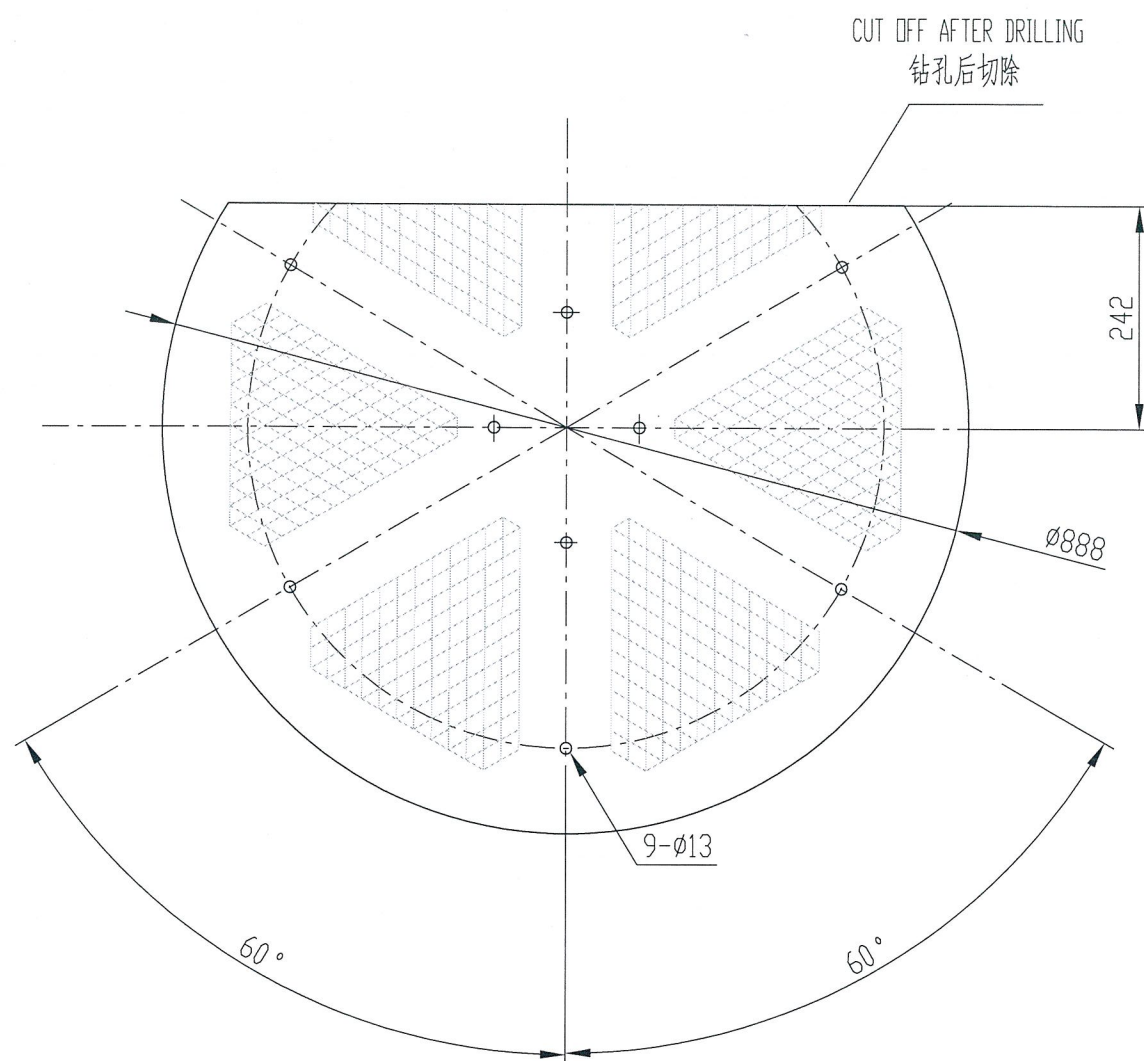


Verifikasi Dokumen
Document Verification

OTHERS
NO. VERIFIKASI: JKKP/2025/01-05/T-01/00881
VERIFICATION NO.
TARIKH: 28/07/2025
DATE



JABATAN KESELAMATAN DAN KESIHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



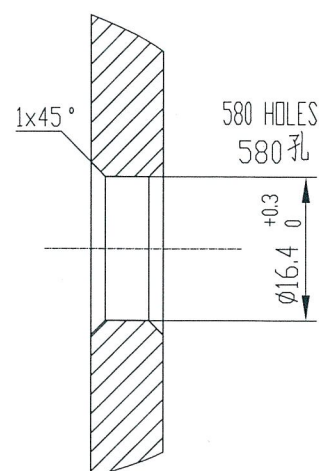
NOTES

技术要求

- 1.THE MARKING OF TUBE HOLES SHALL FOLLOW THE DETAIL MARKING OF TUBE SHEET MARKING. EACH TUBE HOLE SHALL BE DRILLED CONCENTRIC WITH THE CORRESPONDING HOLE ON TUBE SHEET.
管孔划线按管板划线详图, 各管孔须与管板上对应管孔同心配钻;
- 2.CLEARLY MARK THE GAP POSITION DURING ASSEMBLY;
缺口位置装配时作出明显标记;
- 3.THE BAFFLE PLATE SHALL BE LEVELED BEFORE DRILLING;
折流板钻孔前应校平;
- 4.AFTER DRILLING, BURRS AROUND TUBE HOLES SHALL BE REMOVED.
钻孔后应除去管孔周边毛刺。

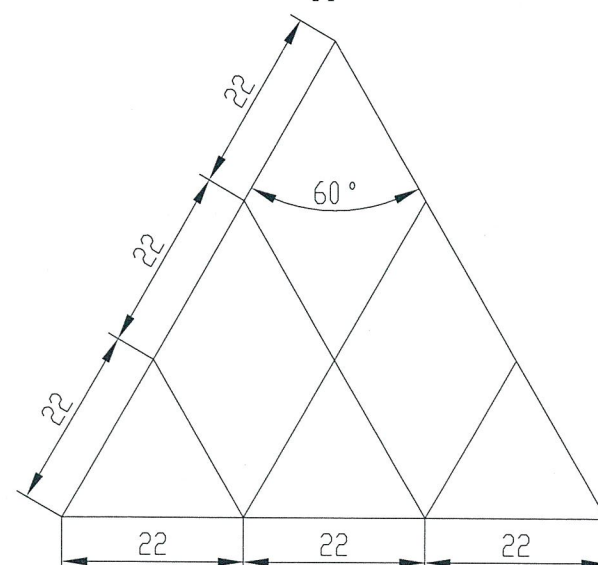
DETAIL DRAWING OF DRILLING
钻孔详图

1:1



DETAIL DRAWING OF TUBE PITCH
孔距详图

1:1



☐ Inspected ☒ reviewed ☐ Witnessed

2024.6.18

						BAFFLE PLATE 折流板				TXNX23-020.03.02	
MARK 标记	QTY 数量	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
DESIGNED BY 设计						Q235B	6	37	1:8		
CHECKED BY 校对											
REVIEWED BY 审核											
APPROVED BY 批准											
						PAGE OF 共 张 第 张					

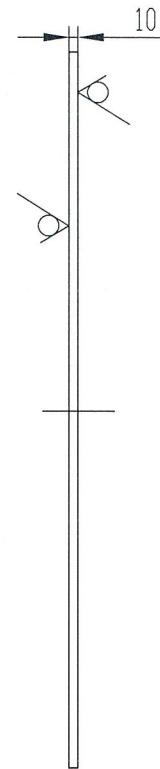
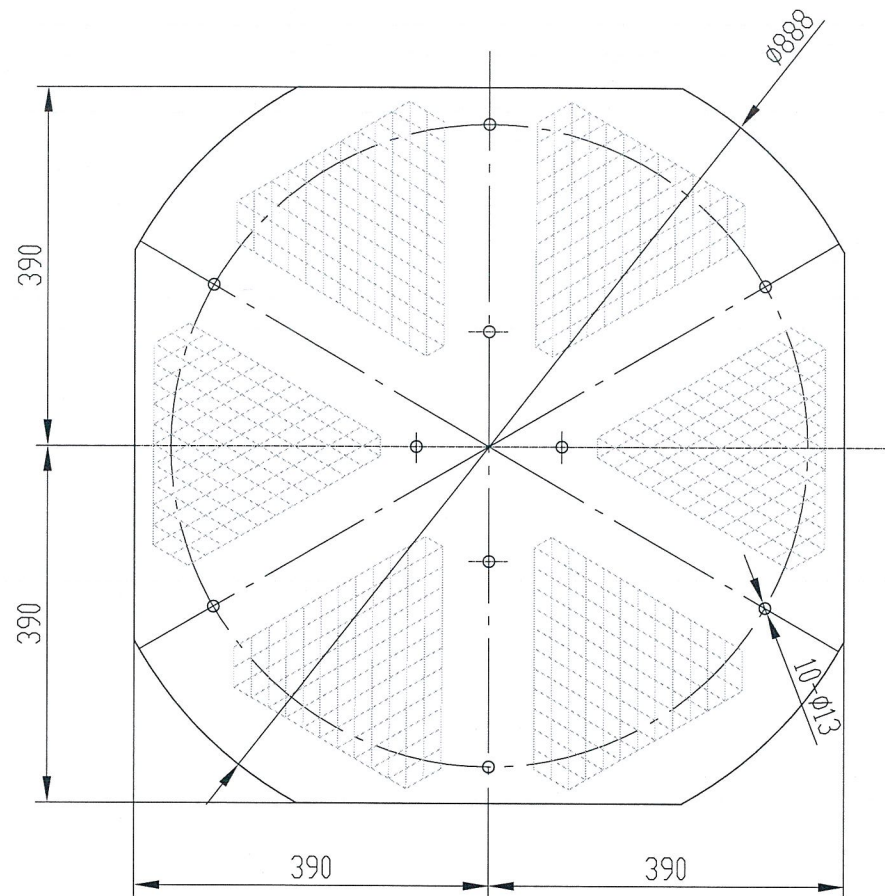


Verifikasi Dokumen
Document Verification

NO. VERIFIKASI: JKKP/2025/01-05/T-01/00881
VERIFICATION NO.
TARIKH: 28/07/2025
DATE

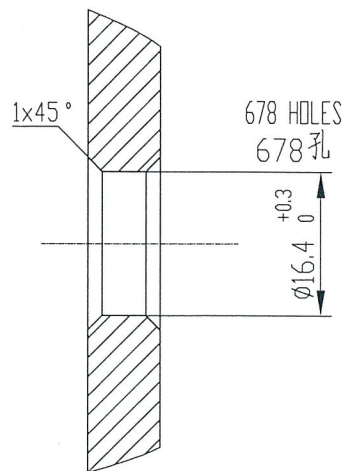
NOTES
技术要求

- 1.THE MARKING OF TUBE HOLES SHALL FOLLOW THE DETAIL IN THE TUBE SHEET MARKING. EACH TUBE HOLE SHALL BE DRILLED CONCENTRIC WITH THE CORRESPONDING HOLE ON TUBE SHEET.
管孔划线按管板划线详图，各管孔须与管板上对应管孔同心配钻；
- 2.CLEARLY MARK THE GAP POSITION DURING ASSEMBLY;
缺口位置装配时作出明显标记；
- 3.THE BAFFLE PLATE SHALL BE LEVELED BEFORE DRILLING;
折流板钻孔前应校平；
- 4.AFTER DRILLING, BURRS AROUND TUBE HOLES SHALL BE REMOVED.
钻孔后应除去管孔周边毛刺。



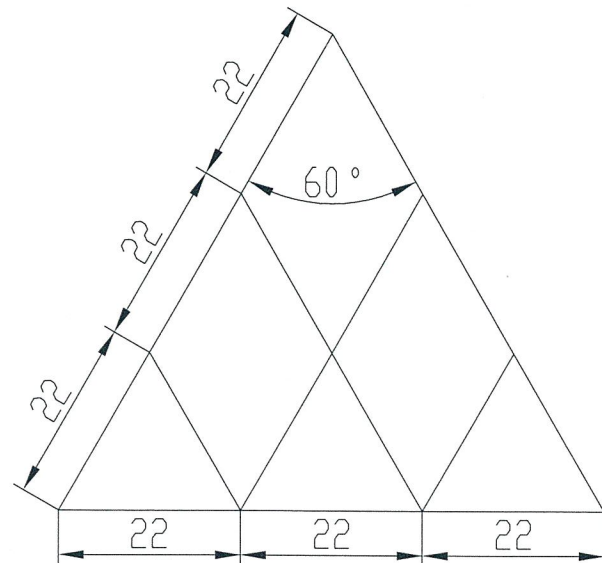
DETAIL DRAWING OF DRILLING
钻孔详图

1:1



DETAIL DRAWING OF TUBE PITCH
孔距详图

1:1



☐ Inspected ☒ Reviewed ☐ Witnessed

Xu Jian
2024.6.18

						BASE PLATE 底板				TXNX23-020.03.05	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日						
DESIGNED BY 设计	陈伟		2024.6.14			MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
CHECKED BY 校对	陈伟		2024.6.14			Q235B	1	30	1:8		
REVIEWED BY 审核						PAGE OF 共 张 第 张					
APPROVED BY 批准											



JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



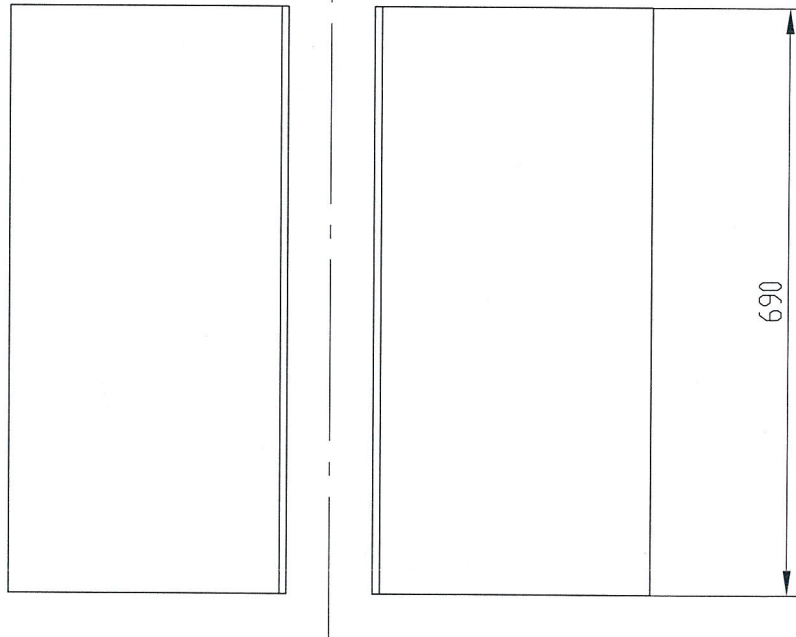
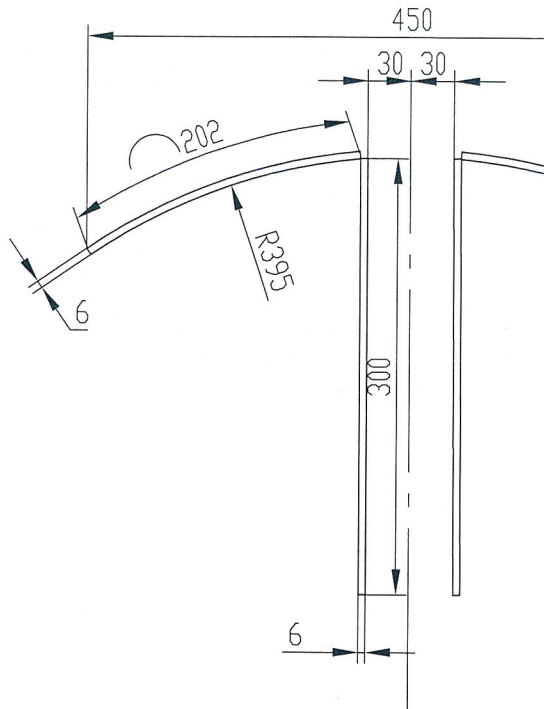
Verifikasi Dokumen
Document Verification

NO. VERIFIKASI: JKKP/2025/01-06/T-01/00881
VERIFICATION NO.
TARIKH: 20/07/2025
DATE



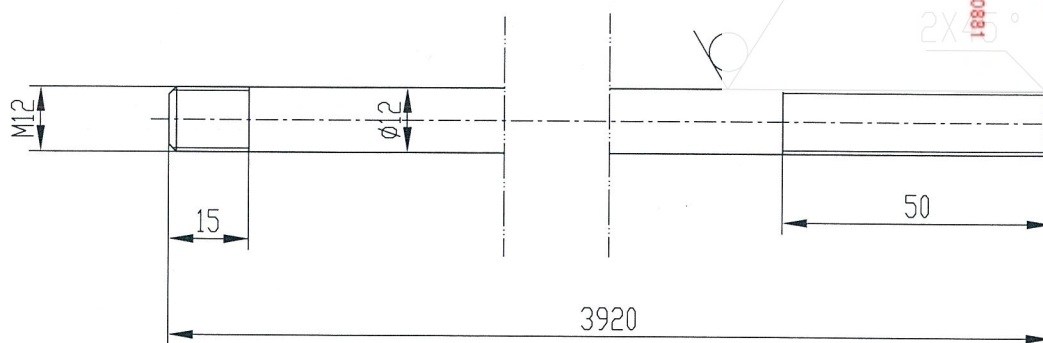
☐ inspected ☒ reviewed ☐ witnessed

Xu Tian
2024.6.18



						STEAM BAFFLE 蒸汽挡板				TXNX23-021.01.01			
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日								
DESIGNED BY 设计	陈		2024.6.14			MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例				
CHECKED BY 校对	陈		2024.6.14			SA-240M 304	1	27.4	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司				
REVIEWED BY 审核													
APPROVED BY 批准													
						PAGE OF 共 张 第 张							

12.5

JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTHVerifikasi Dokumen
Document VerificationNO. VERIFIKASI: JKKP/2025/01-06/T-01/00881
VERIFICATION NO.
TARIKH: 28/07/2025
DATEALIGNMENT IS REQUIRED BEFORE MOUNTING
加工螺孔前对孔

TÜV Rheinland®

☐ inspected ☒ reviewed ☐ witnessedX. Tan.
2024.6.18

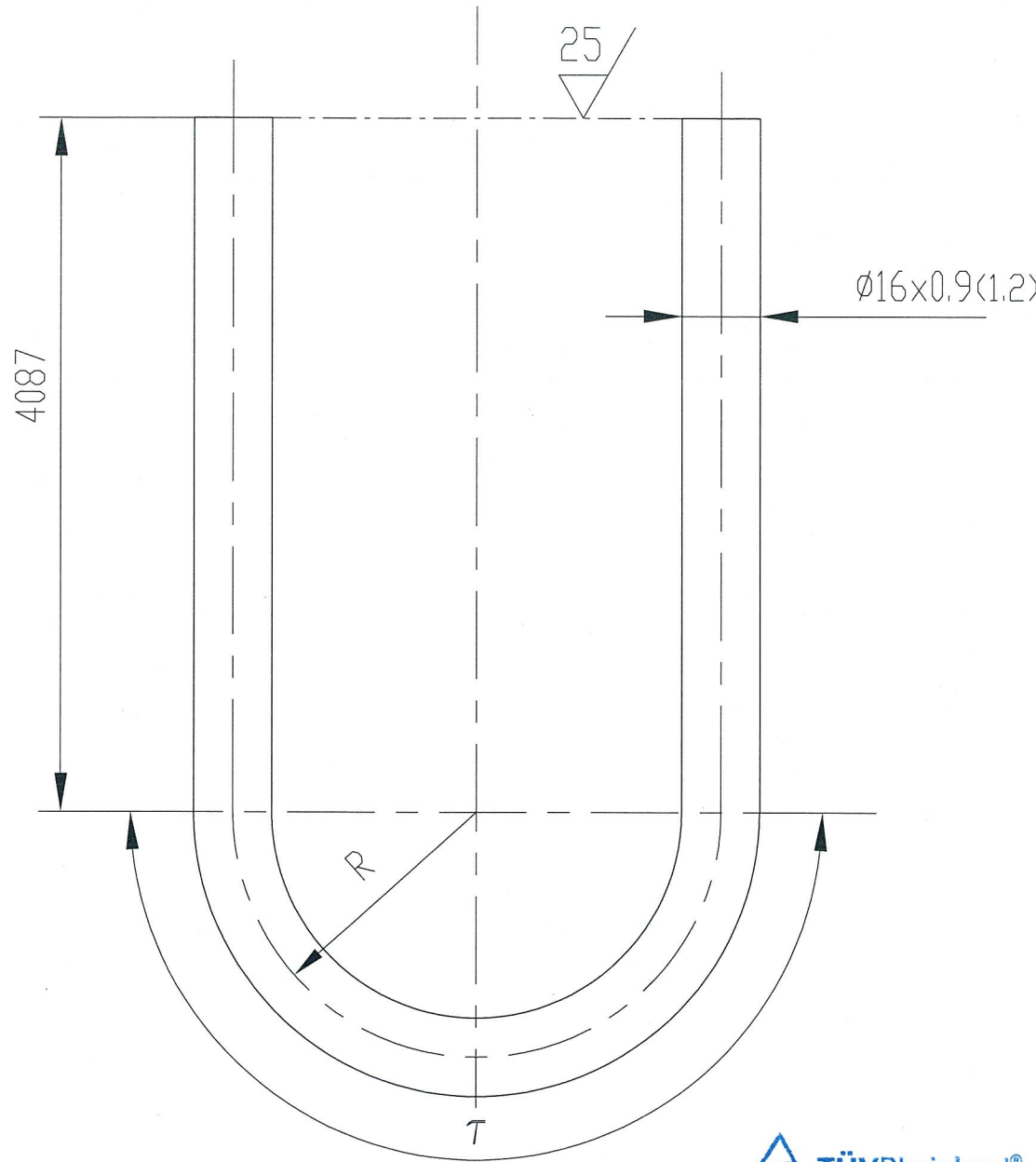
						TIE ROD 拉 杆				TXNX23-021.01.02	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日						
DESIGNED BY 设计	陈伟		2024.6.14			MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例		
CHECKED BY 校对	徐明		2024.6.14			Q235B	10	3.48	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司		
REVIEWED BY 审核											
APPROVED BY 批准											
PAGE OF 共 张 第 张											



Verifikasi Dokumen
Document Verification

TABLE OF SIZE AND QUANTITY
尺寸和数量表

NO.	R mm	T mm	TOTAL LENGTH OF TUBE/PC 管子全长/根 mm	TUBE QUANTITY (PC) 管子数量 (根)	TOTAL WEIGHT 总重 (kg)	TOTAL LENGTH 管子总长 mm
1	50	157	8331	36		
2	69	217	8391	39		
3	88	277	8451	39		
4	107	336	8510	36		
5	126	396	8570	33		
6	145	456	8630	30		
7	164	515	8689	27		
8	183	575	8749	24		
9	202	635	8809	21		
10	222	697	8871	18		
11	241	757	8931	15		
12	260	817	8991	12		
13	279	877	9051	9		
TOTAL 合计				339	985	2917343



TÜVRheinland®
☐ Inspected ☒ Reviewed ☐ Witnessed
Lu Jian
2024.6.18

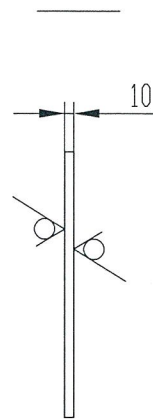
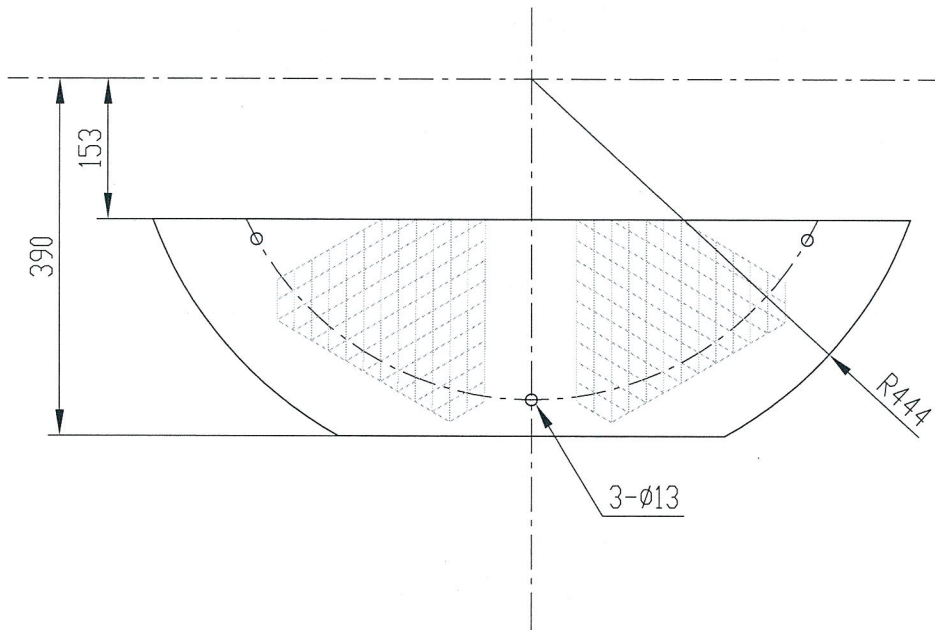
NOTES
技术要求

- 1.TUBES SHALL BE BENT AS A WHOLE. SPLICED WELDING IS NOT ALLOWED;
换热管须整管弯制而成，不得拼焊；
- 2.AFTER BENDING AND FORMING OF TUBES, MAKE HYDROTEST AS PER PIECE WITH A PRESSURE OF 2.75MPa.
KEEP THE PREESURE FOR 10 SECONDS. AFTER THE TEST IS ACCEPTABLE, CLEAN THE WATER STAINS INSIDE TUBES. WHEN THIS REQUIREMENT CANNOT BE MET, THE CHLORIDE ION CONTENT IN WATER SHALL BE CONTROLLED TO NOT EXCEED 25MG/L;
管子弯曲成形后应每根进行水压试验，压力为2.75MPa，保压10S，合格后应将管内水渍处理干净。当无法达到这一要求时，应控制水的氯离子含量不超过25mg/L；
- 3.THE TECHNICAL REQUIREMENTS FOR TUBE PURCHASING SHALL COMPLY WITH ASME SA-688M TP304 STANDARD;
管子订货技术要求按ASME SA-688M TP304标准；
- 4.THE SPECIFICATIONS OF R50 AND R69 TUBES ARE Ø16X1.2.
R50、R69换热管规格为Ø16x1.2。

						TUBE 换热管				TXNX23-021.01.03	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
DESIGNED BY 设计	陈伟		2024.6.14								
CHECKED BY 校对	陈伟		2024.6.14		SA688 TP304	339	985	1:6			
REVIEWED BY 审核											
APPROVED BY 批准						PAGE OF 共 张 第 张					



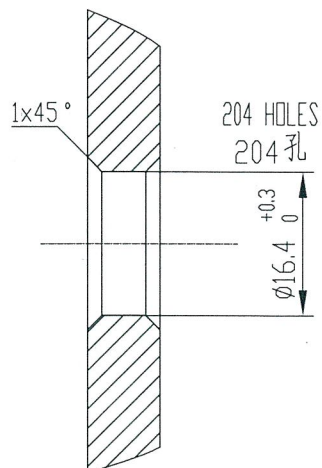
NO. VERIFIKASI: JKKP/2025/01-05/T-01/00881
VERIFICATION NO.
TARIKH: 28/07/2025
DATE



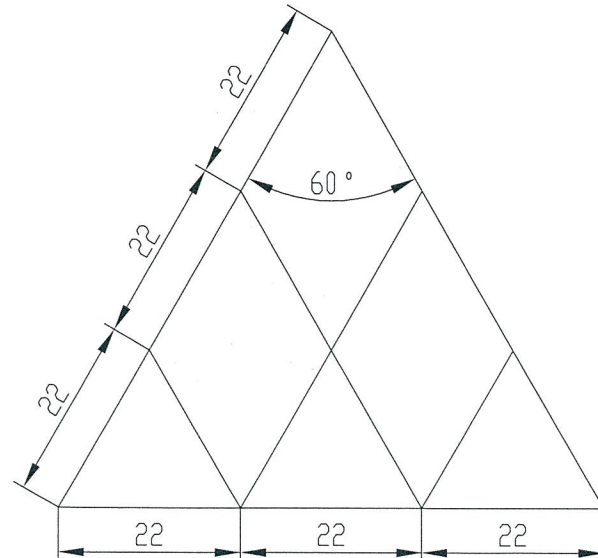
NOTES 技术要求

- 1.THE MARKING OF TUBE HOLES SHALL FOLLOW THE DETAIL DRAWING OF TUBE SHEET MARKING. EACH TUBE HOLE SHALL BE DRILLED CONCENTRIC WITH THE CORRESPONDING HOLE ON TUBE SHEET.
管孔划线按管板划线详图，各管孔须与管板上对应管孔同心配钻；
- 2.CLEARLY MARK THE GAP POSITION DURING ASSEMBLY;
缺口位置装配时作出明显标记；
- 3.THE BAFFLE PLATE SHALL BE LEVELED BEFORE DRILLING;
折流板钻孔前应校平；
- 4.AFTER DRILLING, BURRS AROUND TUBE HOLES SHALL BE REMOVED.
钻孔后应除去管孔周边毛刺。

DETAIL DRAWING OF DRILLING
钻孔详图
1:1



DETAIL DRAWING OF TUBE PITCH
孔距详图
1:1



TÜVRheinland
X. Jia
2024.6.18

						PASS PARTITION 隔板				TXNX23-021.01.04	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司	
DESIGNED BY 设计						Q235B	1	12	1:8		
CHECKED BY 校对											
REVIEWED BY 审核											
APPROVED BY 批准										PAGE OF 共 张 第 张	

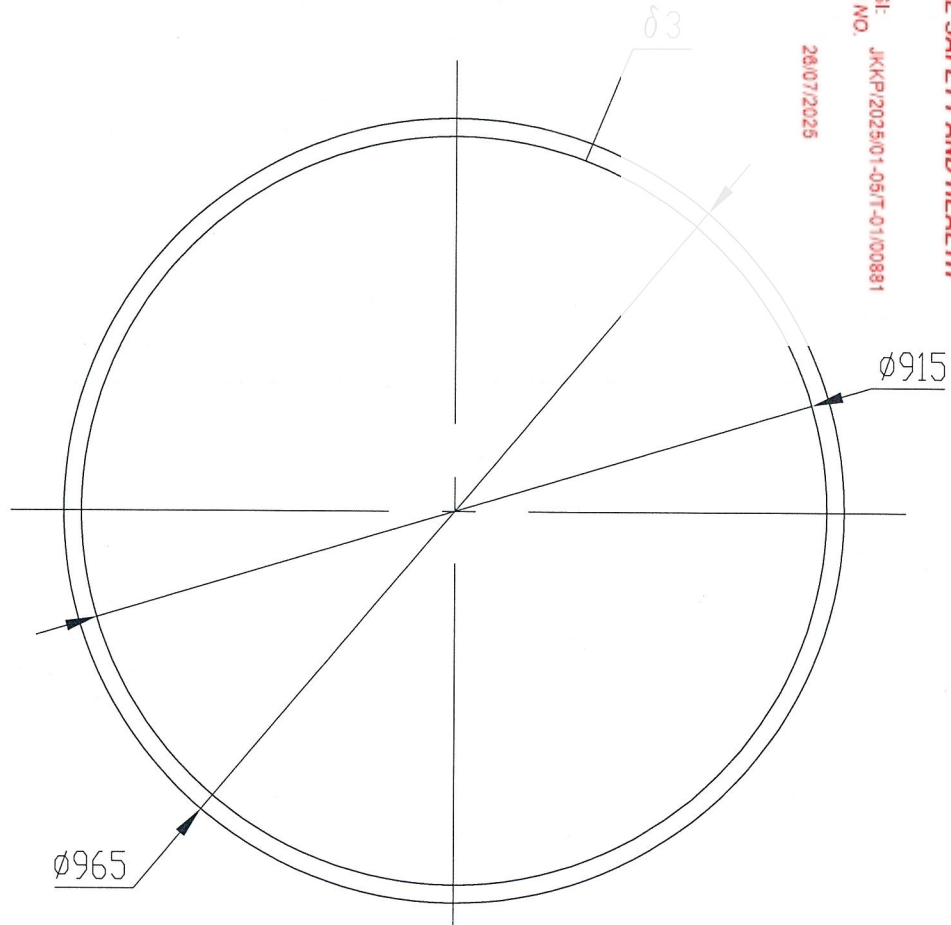


JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



Verifikasi Dokumen
Document Verification

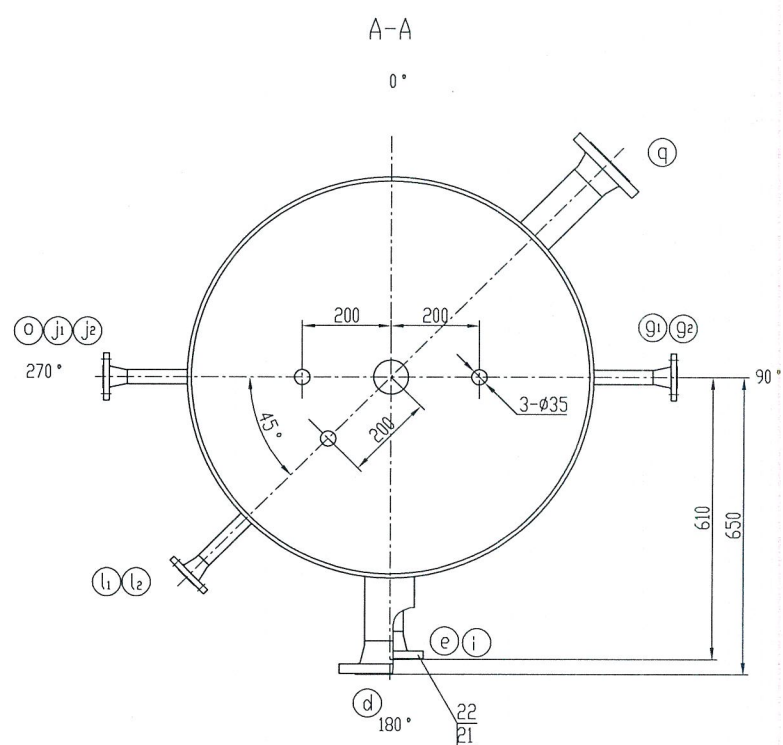
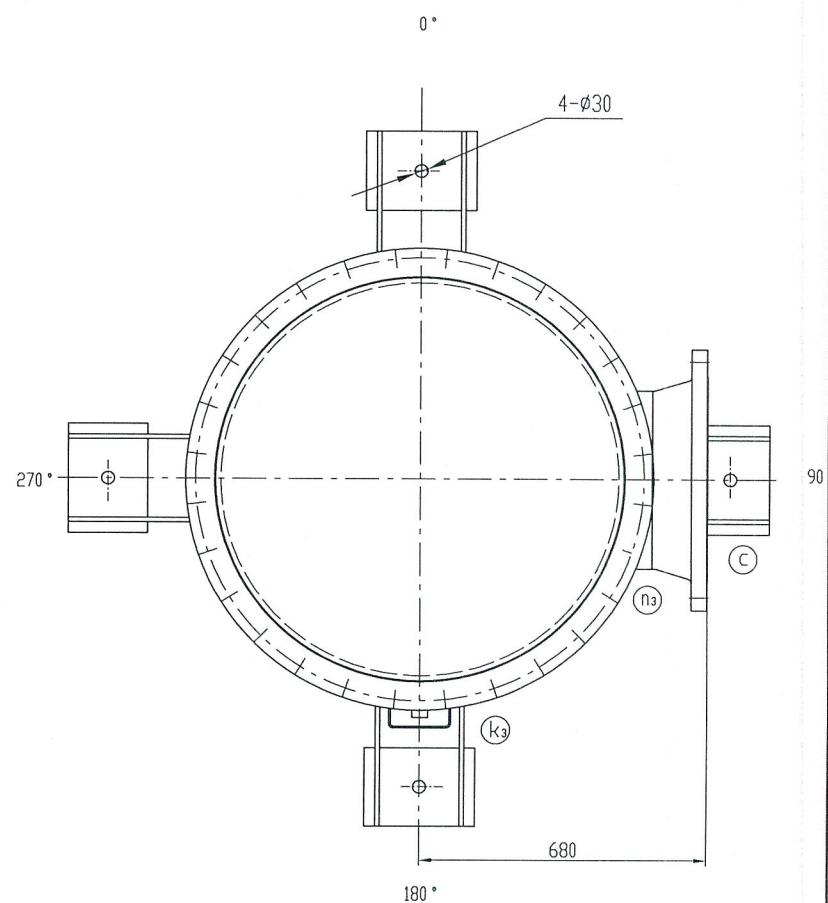
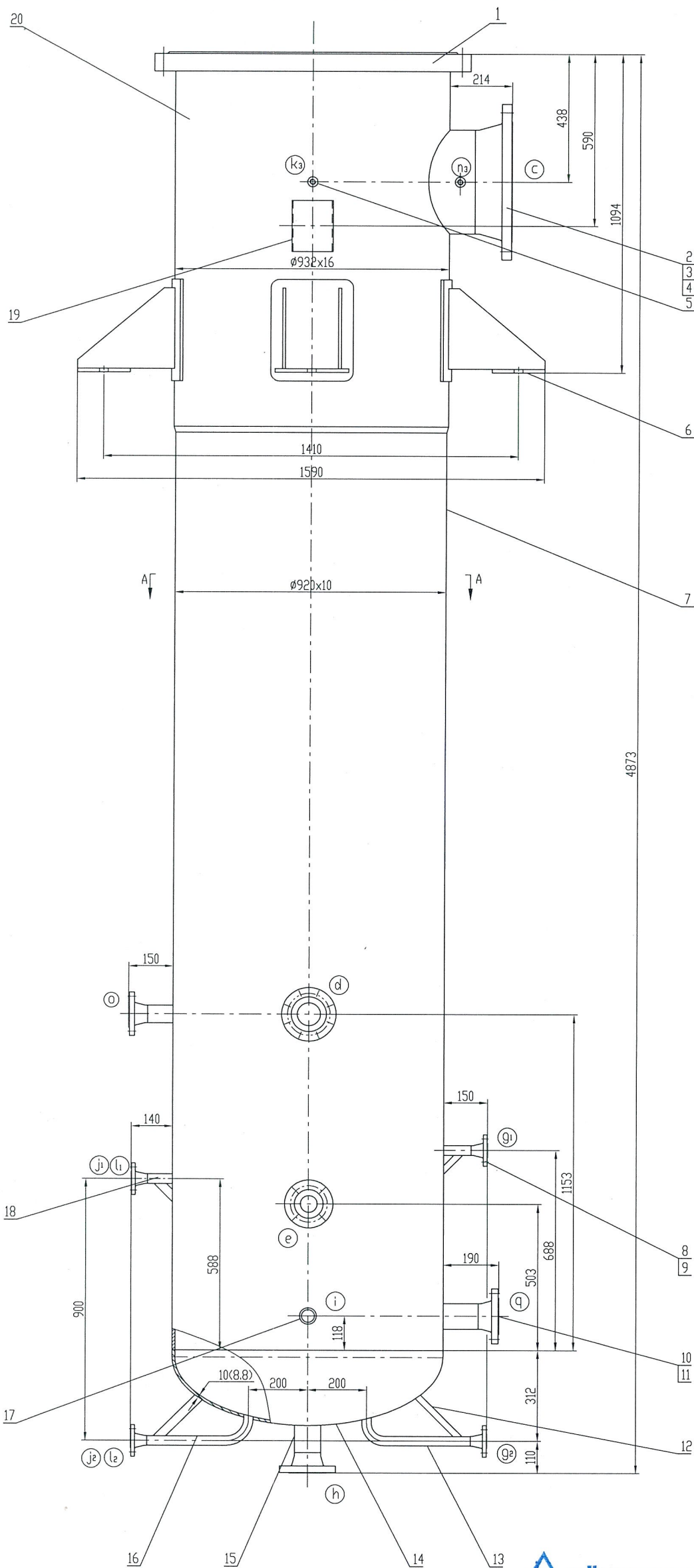
NO. VERIFIKASI: JKKP/2025/01-06/IT-01/00881
VERIFICATION NO.
TARIKH: 26/07/2025
DATE



☐ Inspected ☒ Reviewed ☐ Witnessed

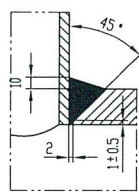
2024.6.18

						SHELL SIDE GASKET				TXNX23-020.04	
						壳侧垫片					
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料		QTY 数量	WEIGHT 重量	PROPORTION 比例	南京汽轮机集团 泰兴宁兴机械有限公司 NANJING TURBINE&ELECTRIC MACHINERY GROUP TAIXING NINGXING MACHINERY CO., LTD.
DESIGNED BY 设计	张伟		2024.6.14		PTFE	1	0.1	1:10			
CHECKED BY 校对	徐时		2024.6.14								
REVIEWED BY 审核											
APPROVED BY 批准					PAGE OF 共 张 第 张						

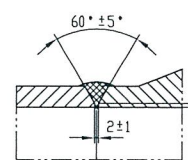


TÜV Rheinland
 Inspected ☒ Reviewed ☒ Witnessed ☒
 2024.6.18

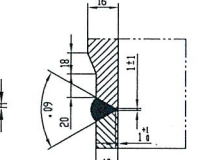
CATEGORY B WELDED JOINT OF NOZZLE
 接管与法兰C类焊接接头
 NO SCALE
 不按比例



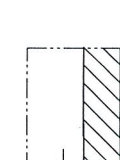
CATEGORY C WELDED JOINT OF NOZZLE-TO-FLANGE
 接管与法兰C类焊接接头
 NO SCALE
 不按比例



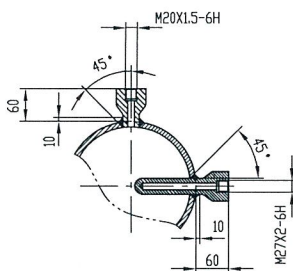
CATEGORY A WELDED JOINT OF THICKNESS PLATE
 A类焊接接头 (不等厚板对接)
 NO SCALE
 不按比例



CATEGORIES A/B WELDED JOINT
 A/B类焊接接头
 NO SCALE
 不按比例



WELDED JOINTS OF THERMOMETER AND PRESSURE GAUGE
 温度计及压力表焊接接头
 NO SCALE
 不按比例



22	STEEL PIPE 规格: 68.3x5.54 L=97	2	SA-106M Gr.B	0.73	1.46	e o
21	FLANGE 法兰: class150 2"	2	SA-105M	3.0	6.0	e o
20	SHELL SECTION 筒节: JN500x16	1	SA-516M Gr.A85	455	H=1257	
19	NAMEPLATE 铭牌	1	ASSEMBLY PARTS	3		
18	STEEL PIPE 规格: 43.4x4.55 L=95	2	SA-106M Gr.B	0.56	1.12	l2 j2
17	STEEL PIPE (INTERNAL EXTENSION 20) 规格: 60x4 L=180x30 (内伸20)	1	SA-182M F304	1.2		i
16	BEND 接管: 43.4x4.55	2	SA-106M Gr.B	1.3	2.6	l2 j2 BORROW
15	NOZZLE 接管: 43.4x4.55	1	SA-106M Gr.B	1.57		h
14	LOWER HEAD 下封头: JN500x16 (0.8R)	1	SA-516M Gr.A85	74.1		BORROW
13	BEND 接管: 43.4x4.55	1	SA-106M Gr.B	1.33		g2 BORROW
12	SUPPORT PLATE 支撑板: 40x10	1组	SA-516M Gr.A85	8.6		LENGTH ADJUSTMENT
11	STEEL PIPE 规格: 68.3x7.62 L=133	2	SA-106M Gr.B	2.03	4.06	q d
10	FLANGE 法兰: class150 3"	3	SA-105M	5.2	15.6	q d h
9	STEEL PIPE 规格: 68.3x5.54 L=105	1	SA-106M Gr.B	0.56		qi

NO. VERIFIKASI: JKKP/2025/01-05/T-01/00881
 TARIKH: 29/07/2025
 VERIFIKASI: 2024.6.18

JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
 DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH

VERIFIKASI DOKUMEN
 文件验证

QR CODE

SHAWANG TUBROTEK ELEC MACHINERY GROUP
 台州宁兴机械有限公司

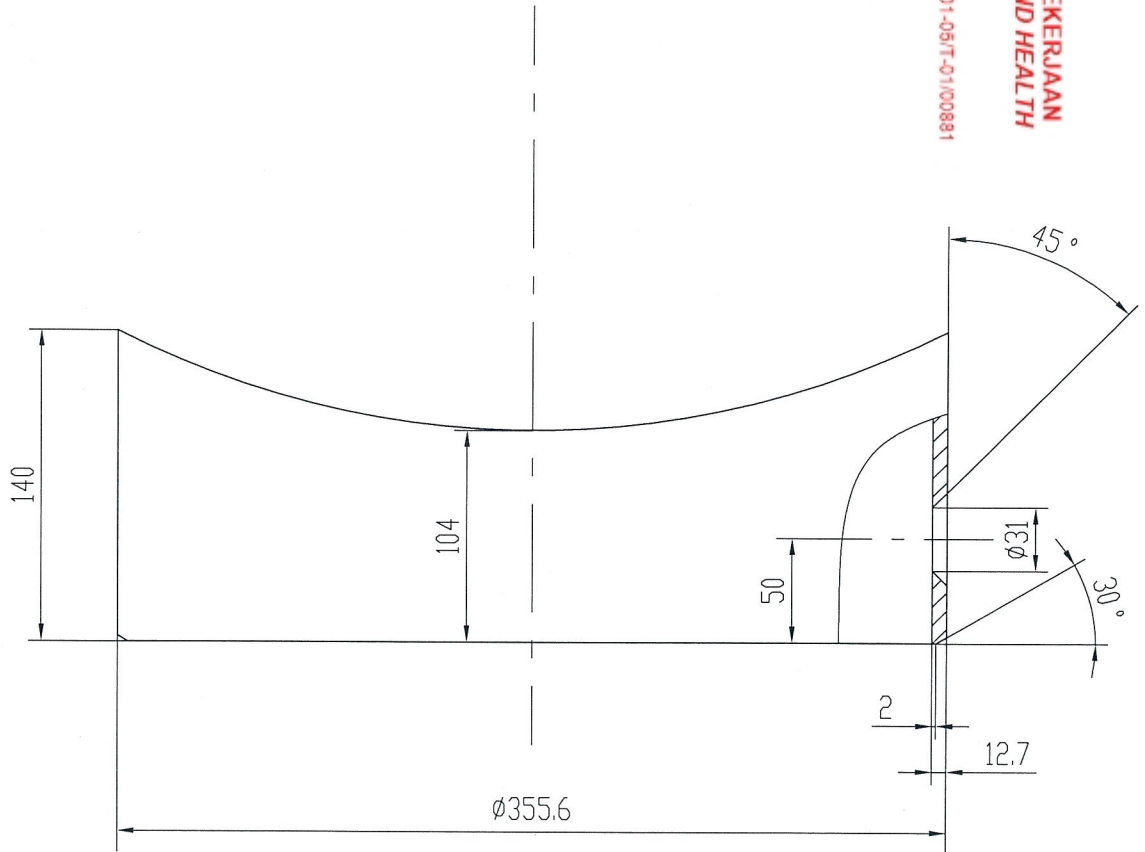


JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



Verifikasi Dokumen
Document Verification

NO. VERIFIKASI: JKKP/2025/01-06/IT-01/00881
VERIFICATION NO.
TARIKH: 28/07/2025
DATE



TÜVRheinland®

☐ inspected ☒ reviewed ☐ witnessed

Xu Jian
2024.6.18

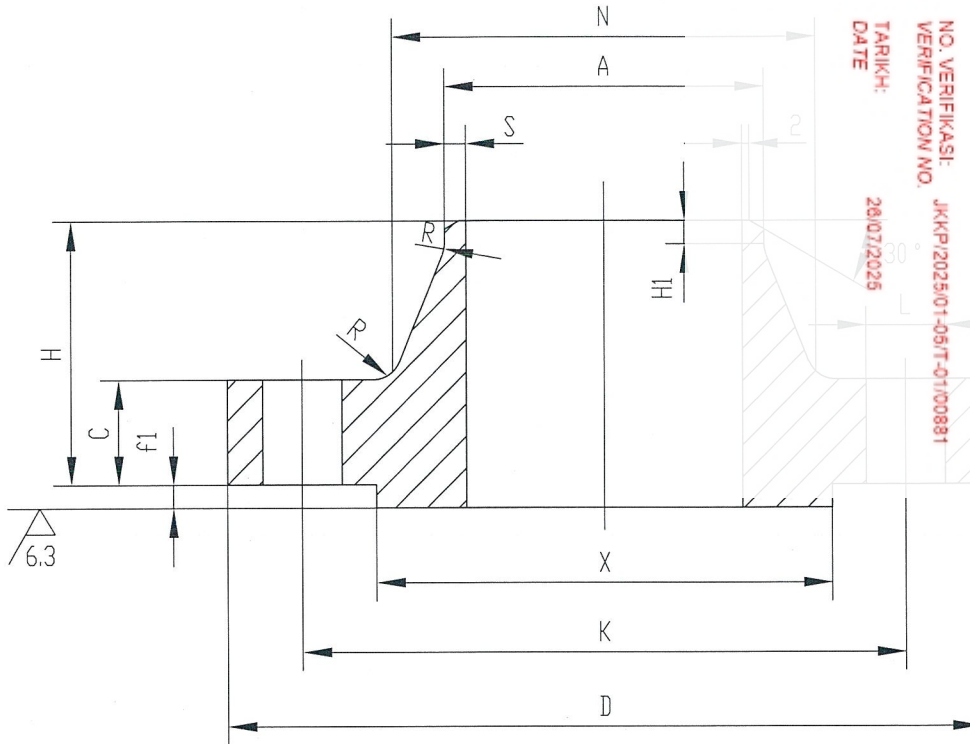
						STEAM INLET PIPE 进汽管				TXNX23-021.02.01					
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日										
DESIGNED BY 设计		陈伟		2024.6.14		MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司					
CHECKED BY 校对		李时		2024.6.14		SA-106M Gr.B	1	7.5	1:2						
REVIEWED BY 审核															
APPROVED BY 批准												PAGE OF 共 张 第 张			



JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH



NO. VERIFIKASI: JKKP/2025/01-06/T-01/00881
VERIFICATION NO.
TARIKH: 20/07/2025
DATE



ASME B16.5 WN RF

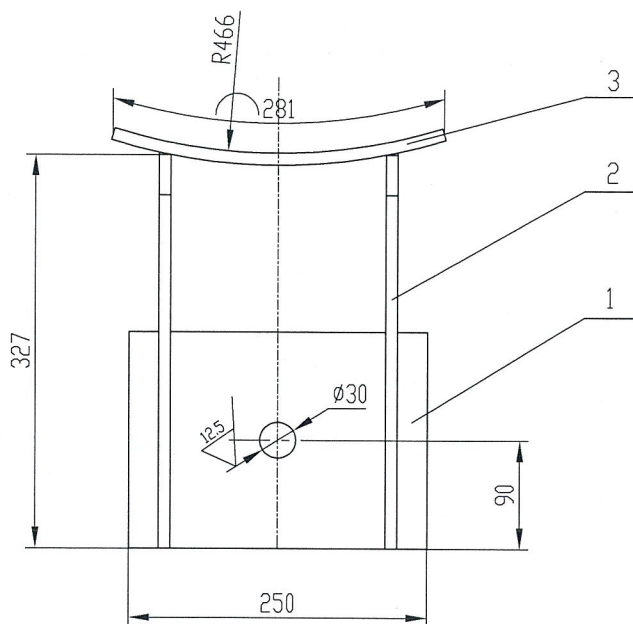
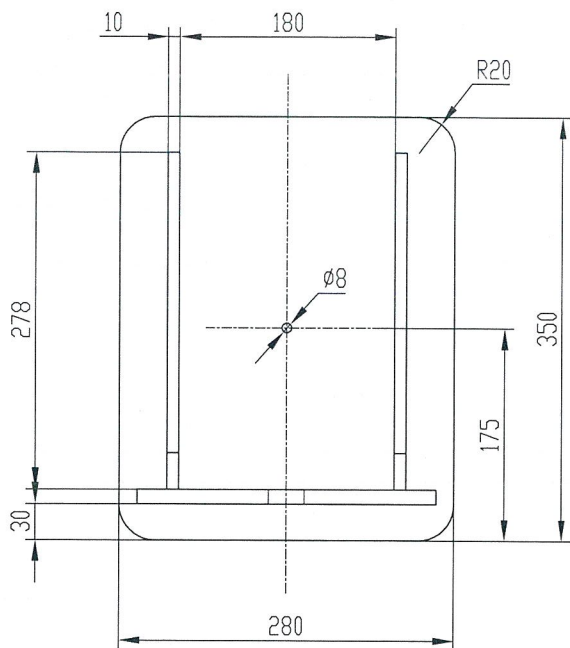
MODEL 型号	A	D	K	L	NO. OF SCREW HOLES 螺孔数	X	f1	C	H	S	N	R	NO. OF FLANGE 法兰数	H1	UNIT WEIGHT 单件重	MATERIAL 材料
class150 1"	33.4	108	79.2	5/8"	4	50.8	1.5	12.7	54	4.55	49	4	6	6	1.0	SA-105M
class150 2"	60.3	152	120.6	3/4"	4	91.9	1.5	17.5	62	5.54	78	8	2	6	3.0	SA-105M
class150 3"	88.9	190	152.4	3/4"	4	127	1.5	22.4	68	7.62	108	10	3	10	5.0	SA-105M
class150 14"	355.6	533	476.2	1-1/8"	12	412.8	1.5	33.3	125	12.7	400	13	1	16	51.7	SA-105M



☐ inspected ☒ reviewed ☐ witnessed

Li. Jian
2024.6.18

						FLANGE 法 兰				TXNX23-021.02.02				
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日	MATERIAL 材料	QTY 数量	WEIGHT 重量	PROPORTION 比例	TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司				
DESIGNED BY 设计	陈伟		2024.6.14											
CHECKED BY 校对	陈伟		2024.6.14											
REVIEWED BY 审核														
APPROVED BY 批准														
						PAGE OF 共 张 第 张								



Verifikasi Dokumen
Document Verification

NO. VERIFIKASI: JKKP/2025/01-06/T-01/00881
VERIFICATION NO.
TARIKH: 20/07/2025
DATE

JABATAN KESELAMATAN DAN KESEHATAN PEKERJAAN
DEPARTMENT OF OCCUPATIONAL SAFETY AND HEALTH

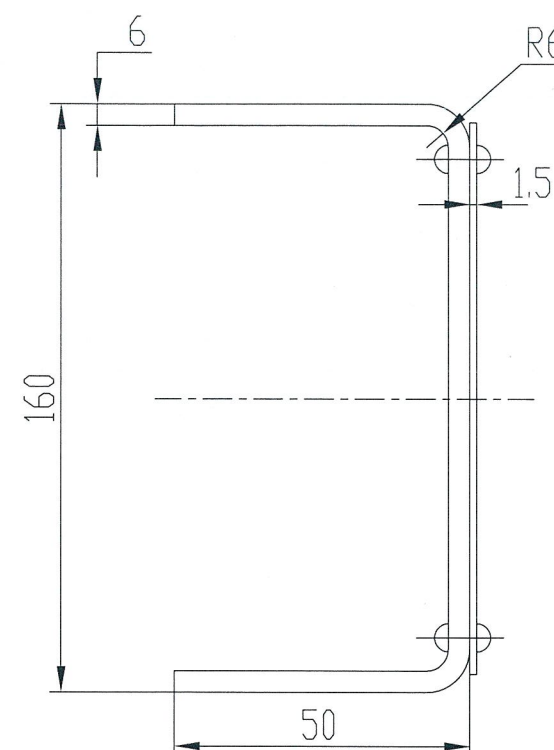
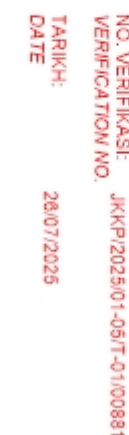


TÜVRheinland®

☐ Inspected ☒ reviewed ☐ witnessed

2024.6.18

3	本图	垫板 $\delta 10$	1	SA-516M Gr.485		7.64	
2	本图	筋板 $\delta 10$	2	Q235B	4.3	8.6	
1	本图	底板 $\delta 12$	1	Q235B		3.45	
PART NO. 件 号	DRAWING NO. OR STANDARD 图号或标准	DESCRIPTION 名 称	QTY 数量	MATERIAL 材 料	UNIT 单	TOTAL 总	NOTE 备 注
					WEIGHT 质量 (KG)		
				EAR LUG 耳式支座		TXNX23-020.05.03	
MARK 标记	QTY 处数	SUBAREA 分区	REVISED DOC. NO. 更改文件号	SIGNATURE 签名	DATE 年月日		
DESIGNED BY 设计	陈伟		2024.6.14		MATERIAL 材料	QTY 数量	WEIGHT 重量
CHECKED BY 校对	徐时		2024.6.14		ASSEMBLY PARTS 组合件	4	19.69
REVIEWED BY 审核					PROPORTION 比例	1:4	
APPROVED BY 批准					PAGE OF 共 张 第 张		
					TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司		



2024-6.18

说明

铭牌由1.5mm的不锈钢制造。

铭牌上的线条、字母和数据框格凸起于板面,且抛光处理,其余部分浸蚀0.1mm深且涂以黑色瓷釉。

所有必须的标记为5mm高。

容器竣工后敲认证标记。

3		BRACKET 铭牌架	1	SA-516M.Gr.485		3.0			
2	ISO 1482	RIVET 铆钉 3.5x 9.5	8	SS 304	0.015	0.06			
1		NAMEPLATE 铭牌 δ=1.5	2	SS 304	0.1	0.2			
PART NO.	DWG. NO. OR STANDARD	DESCRIPTION	QTY	MATERIAL	UNIT	TOTAL	NOTE		
件 号	图号或标准	名 称	数量	材 料	单	总	备 注		
					WEIGHT				
					质量(kg)				
MARK	QTY	SUBAREA	REVISED DOC. NO.	SIGNATURE	DATE	REV. NO.	QTY	WEIGHT	PROPORTI
标记	处数	分区	更改文件号	签名	年月日	版本	数量	重量	比例
DESIGNED BY									
设计									
CHECKED BY									
校对									
REVIEWED BY									
审核									
APPROVED BY									
批准									
PAGE 4 OF 4 共 4 张 第 4 张						TAIXING NINGXING MACHINERY CO., LTD. 泰兴宁兴机械有限公司			